

An Enhanced REGCN Framework with Simplified Recurrent and Attention-Gated Temporal Modules for Stock Price Prediction

HIMANSHU JAIN¹, GURSIMARJEET SINGH², SWEETI SAH¹

¹Department of Computer Science and Engineering, National Institute of Technology Kurukshetra, India (e-mail: himanshujain102004@gmail.com)

²Department of Mechanical Engineering, National Institute of Technology Kurukshetra, India (e-mail: gursimarjeet098@gmail.com)

Corresponding author: Sweeti Sah (e-mail: your@email.com).

ABSTRACT Stock price forecasting remains a challenging task due to nonlinear temporal dynamics, structural dependencies among transaction attributes, and the presence of market noise. Graph-based deep learning models, such as the Relational Embedding Graph Convolutional Network (REGCN), have demonstrated the effectiveness of integrating relational modeling with temporal learning. However, the influence of temporal modeling design under strictly controlled graph construction and training settings has not been thoroughly investigated. This study revisits the REGCN framework and proposes two alternative temporal modeling architectures within an identical preprocessing and graph construction pipeline. The first model introduces a simplified recurrent structure that removes redundant gating components to improve parameter efficiency and training stability. The second replaces recurrent modeling with a temporal self-attention mechanism equipped with adaptive gating, enabling parallel modeling of long-range dependencies. To address non-stationarity in financial time series, Variational Mode Decomposition (VMD) is applied in a per-attribute manner, with decomposition parameters optimized on the training set to prevent data leakage. A trend-aware composite loss function is employed to jointly optimize numerical accuracy and directional consistency. During evaluation, a margin-based criterion is used for the proposed models to reduce the impact of minor price fluctuations on trend assessment, while the baseline REGCN is evaluated under its original protocol to ensure fairness. Experiments conducted on benchmark datasets and an additional NASDAQ dataset demonstrate that, under strictly controlled experimental conditions, the proposed architectures achieve competitive numerical performance and stable trend prediction behavior. The results indicate that careful temporal modeling design within a unified graph framework can improve robustness without altering the underlying relational modeling strategy.

INDEX TERMS Stock Price Forecasting, Graph Neural Networks, Relational Graph Convolution, Variational Mode Decomposition, Temporal Self-Attention, Financial time-series

I. INTRODUCTION

Financial time-series forecasting remains one of the most challenging tasks in quantitative finance and machine learning. Stock prices exhibit strong nonlinearity, non-stationarity, volatility clustering, and substantial noise contamination [1], [2]. These characteristics make accurate prediction inherently difficult. Nevertheless, reliable forecasting models are highly valuable for portfolio allocation, algorithmic trading, and risk management. Even incremental improvements in predictive stability or directional accuracy may yield meaningful practical benefits.

Traditional approaches to stock price forecasting are pri-

marily based on statistical time-series models. Methods such as Autoregressive Integrated Moving Average (ARIMA) and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) have been widely adopted due to their solid theoretical foundation and interpretability [3]. While effective for modeling linear dependencies and conditional variance, these models rely on assumptions that are often violated in real financial markets, particularly under nonlinear dynamics and structural breaks.

Recent advances in artificial intelligence, especially deep learning, have significantly expanded the modeling capacity available for financial forecasting. Architectures such as

Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRUs), and transformer-based models have demonstrated strong capability in capturing nonlinear temporal dependencies [4]–[7]. Owing to their flexible representation learning ability, deep learning approaches have increasingly outperformed classical statistical models in multivariate forecasting settings [8].

Despite these advances, many existing deep learning models treat financial time series primarily as independent temporal sequences and focus predominantly on temporal feature extraction. However, transaction attributes within a stock—such as open, high, low, close, and volume—exhibit structured dependencies that can provide valuable inductive bias. Modeling such attribute-level relationships may enhance representation learning and improve forecasting stability.

Graph neural networks (GNNs) offer a principled framework for capturing structured dependencies. Graph Convolutional Networks (GCNs) extend convolution operations to non-Euclidean domains by jointly learning from node features and graph topology [10]. In financial forecasting, transaction attributes can be modeled as nodes, while statistical or distance-based measures define relational edges. By propagating information across this structured graph, GCNs enable relation-aware feature extraction beyond purely sequential modeling. Several recent studies have demonstrated the effectiveness of graph-based architectures in financial time-series prediction [11], [12].

Another important challenge in stock forecasting arises from non-stationarity and multi-scale temporal fluctuations. Market prices are influenced by heterogeneous factors operating at different frequencies, including long-term trends and short-term volatility. Signal decomposition techniques have therefore been introduced to separate complex time series into components corresponding to distinct frequency bands. Variational Mode Decomposition (VMD) is particularly attractive due to its variational formulation and noise robustness [13]. By decomposing financial signals into intrinsic mode functions (IMFs), VMD facilitates multi-scale modeling and has shown promising results when combined with deep learning architectures [14], [15].

In addition to numerical precision, directional consistency is often of practical importance in financial applications. Correctly predicting the direction of price movement can be more valuable than minimizing point-wise regression error. However, many forecasting frameworks optimize regression-based objectives while evaluating trend accuracy only at inference time [16]. This discrepancy between training objectives and evaluation criteria may affect directional stability. Recent research has therefore explored incorporating directional considerations into the optimization process [17].

To integrate multi-scale decomposition, relational modeling, and temporal learning, Ma and Yuan proposed the Relational Embedded Graph Convolutional Network (REGCN) [18]. REGCN applies VMD to transaction attributes, constructs multi-relational graphs to model attribute dependen-

cies, and employs recurrent units for temporal modeling. A trend-aware regularization term is further incorporated into the loss function to balance numerical accuracy and directional correctness.

While REGCN demonstrates strong empirical performance, its recurrent structure and repeated graph operations may limit computational efficiency, particularly when applied to multiple decomposed components. Furthermore, alternative temporal modeling strategies may provide complementary advantages in capturing long-range dependencies.

Motivated by these considerations, this paper proposes two graph-based forecasting architectures developed under the same preprocessing, graph construction, and evaluation protocol as REGCN to ensure fair comparison. The first model introduces a refined recurrent architecture that simplifies the gating mechanism and incorporates implementation-level optimizations such as adjacency caching and efficient graph execution. The second model replaces recurrence with a temporal self-attention module enhanced by a gating mechanism, enabling parallel modeling of temporal dependencies.

All models are evaluated following the original REGCN experimental setup on benchmark datasets, and the same pipeline is extended to an additional NASDAQ dataset to examine robustness under different market conditions. Experimental results indicate that the proposed architectures achieve competitive predictive performance while improving computational efficiency.

The remainder of this paper is organized as follows. Section II reviews related work on statistical, deep learning, and graph-based stock prediction models. Section III describes the proposed methodology in detail. Section IV presents the datasets and experimental setup. Section V discusses experimental results and ablation studies. Section VI concludes the paper and outlines future research directions.

II. LITERATURE REVIEW

This section reviews representative studies on stock price forecasting, including statistical approaches, deep learning-based methods, graph-based frameworks, and decomposition-driven and trend-aware models. The objective is to summarize existing progress while identifying methodological gaps that motivate the present study.

A. STATISTICAL MODELS FOR STOCK PRICE PREDICTION

Early research on financial forecasting relied heavily on statistical and econometric models. Linear time-series models such as Autoregressive Integrated Moving Average (ARIMA) were widely adopted due to their interpretability and solid theoretical foundation. These models assume linear dynamics and stationarity, assumptions that are often violated in practical financial markets characterized by structural breaks and nonlinear behavior.

To model time-varying volatility, Engle introduced the Autoregressive Conditional Heteroskedasticity (ARCH) model and its generalized form, GARCH [3]. GARCH-based models effectively capture volatility clustering and conditional

variance dynamics. Nevertheless, their ability to describe nonlinear price evolution and complex cross-variable interactions remains limited. As a result, although statistical models remain valuable for volatility modeling and risk estimation, their direct predictive performance on stock price levels is often constrained in highly nonlinear environments.

B. DEEP LEARNING-BASED STOCK FORECASTING MODELS

With advances in machine learning, nonlinear predictive models have become increasingly prominent in financial forecasting. Tay and Cao [1] demonstrated that support vector machines outperform traditional linear models in certain financial prediction tasks, highlighting the importance of nonlinear representation learning.

Subsequently, deep learning models have been extensively explored. Convolutional Neural Networks (CNNs) have been applied to capture local temporal patterns in stock price sequences [4]. While CNNs are effective in extracting short-term features, their ability to model long-range dependencies is inherently limited.

Recurrent neural networks, particularly Long Short-Term Memory (LSTM) networks, are widely used due to their capability to model long-term temporal dependencies [5]. Gated Recurrent Units (GRUs), which reduce the number of gating mechanisms, provide improved computational efficiency while maintaining competitive forecasting performance [6]. Despite these advantages, recurrent models require sequential processing, which limits parallelization and may increase training time when handling long input sequences.

More recently, attention-based and Transformer architectures have been introduced to address these limitations. Li et al. [7] proposed a Transformer-based time-series model that captures global temporal dependencies through self-attention mechanisms. A recent survey by Liu et al. [8] reports growing adoption of Transformer architectures in financial forecasting. Nonetheless, many deep learning models treat each stock independently and focus primarily on temporal modeling without explicitly incorporating relational dependencies among assets.

C. MULTIVARIATE AND GRAPH-BASED FORECASTING APPROACHES

Financial assets exhibit significant cross-sectional correlations arising from industry structures, macroeconomic exposure, and market sentiment. Lin et al. [9] showed that incorporating multiple correlated variables improves forecasting performance compared to univariate approaches. However, conventional multivariate models typically rely on implicit correlation modeling and do not explicitly encode structured relationships.

Graph Neural Networks (GNNs) offer a principled framework for modeling structured dependencies. Kipf and Welling [10] introduced Graph Convolutional Networks (GCNs), which extend convolution operations to graph-structured data by aggregating information from neighboring nodes. This

framework has been applied to financial forecasting scenarios where stocks are represented as nodes and inter-stock relationships as edges.

Wang et al. [11] proposed a Temporal Graph Convolutional Network (T-GCN) that combines graph convolution with recurrent modeling to jointly capture spatial and temporal dependencies. Zhou et al. [12] further introduced dynamic graph neural networks to account for evolving relationships over time. Although graph-based models demonstrate improved modeling flexibility, their effectiveness depends on graph construction strategies and may introduce additional computational complexity.

D. SIGNAL DECOMPOSITION AND TREND-AWARE FORECASTING

Financial time series are influenced by multiple temporal scales, including long-term economic cycles and short-term speculative fluctuations. Signal decomposition techniques aim to separate these components to facilitate more effective learning. Variational Mode Decomposition (VMD), proposed by Dragomiretskiy and Zosso [13], formulates signal decomposition as a constrained variational optimization problem, enabling robust separation of intrinsic mode functions.

Recent studies have integrated VMD with deep learning architectures for financial forecasting. Wang et al. [14] demonstrated that combining VMD with neural networks improves predictive stability by isolating multi-scale temporal patterns. Ali and Khan [15] explored decomposition-driven forecasting using CEEMDAN and deep models, reporting improved robustness under noisy conditions. However, many decomposition-based frameworks process components independently without incorporating relational structures among assets.

Beyond numerical accuracy, directional prediction is particularly relevant in practical trading applications. Bollerslev et al. [2] emphasized the role of directional information in financial risk assessment. In many forecasting models, trend accuracy is evaluated only after training using regression-based objectives.

To address this gap, Zhang et al. [16] proposed trend-focused neural models that emphasize directional learning. Sun et al. [17] incorporated directional accuracy into the optimization objective to enhance robustness to market noise. These studies indicate that aligning training objectives with practical evaluation criteria may improve real-world applicability.

Building upon graph-based modeling, signal decomposition, and trend-aware optimization, Ma and Yuan proposed the Relational Embedded Graph Convolutional Network (REGCN) [18]. REGCN integrates VMD-based decomposition, multi-relational graph convolution, and gated recurrent units to capture spatio-temporal dependencies in transaction attribute data. A trend regularization term is incorporated into the loss function to balance numerical precision and directional consistency.

While REGCN demonstrates strong empirical results, its recurrent structure and repeated graph operations across decomposed components may affect computational efficiency when scaling to larger datasets. These considerations motivate further architectural refinement and implementation-level optimization.

In summary, existing research highlights the importance of nonlinear modeling, relational learning, signal decomposition, and directional awareness in stock price forecasting. Nevertheless, balancing predictive performance with computational efficiency and scalability remains an open challenge. The present study builds upon the REGCN framework and investigates two alternative temporal modeling strategies under an identical preprocessing and evaluation protocol, with the objective of improving efficiency while maintaining predictive capability.

III. RESEARCH METHODOLOGY

A. PROBLEM FORMULATION

Stock price prediction is formulated as a multivariate time series forecasting problem with structured relational dependencies among transaction attributes. Financial markets exhibit complex temporal dynamics, nonlinear correlations, and significant noise components, requiring joint modeling of both spatial (inter-attribute) and temporal dependencies.

Let the daily transaction state of a stock at time step t be represented as a multivariate feature vector:

$$\mathbf{s}_t \in \mathbb{R}^{N \times 1}, \quad (1)$$

where N denotes the number of transaction attributes (e.g., open, high, low, close, volume).

Given a historical observation window of length L , the input sequence is defined as

$$\mathbf{S}_t = (\mathbf{s}_{t-L+1}, \mathbf{s}_{t-L+2}, \dots, \mathbf{s}_t), \quad (2)$$

and the forecasting objective is to estimate the closing price at the next time step:

$$\hat{y}_{t+1} = \hat{C}_{t+1}^{\text{close}}. \quad (3)$$

a: Multi-Scale Decomposition.

To mitigate non-stationarity and noise effects, each transaction attribute is decomposed into K intrinsic mode functions (IMFs) using Variational Mode Decomposition (VMD). For attribute $x_t^{(n)}$, the decomposition is expressed as:

$$x_t^{(n)} = \sum_{k=1}^K u_t^{(n,k)}, \quad (4)$$

where $u_t^{(n,k)}$ denotes the k -th decomposed component.

Prediction is performed independently on each IMF component, and the final forecast is obtained via aggregation:

$$\hat{y}_{t+1} = \sum_{k=1}^K \hat{y}_{t+1}^{(k)}, \quad (5)$$

b: Relational Graph Modeling.

To capture structural dependencies among transaction attributes, the attribute space is modeled using a set of relational graphs:

$$\mathcal{G} = \{G_1, G_2, \dots, G_m\}, \quad (6)$$

where each graph $G_i = (V, E_i)$ represents a specific dependency type. Here, each node $v \in V$ corresponds to a transaction attribute, and weighted edges encode attribute-wise relationships computed using statistical or distance-based measures (e.g., Pearson correlation, Spearman correlation, and Dynamic Time Warping).

Under this formulation, the forecasting task aims to learn a mapping function:

$$\hat{y}_{t+1} = f(\mathbf{S}_t, \mathcal{G}), \quad (7)$$

where $f(\cdot)$ denotes a graph-based temporal prediction model integrating multi-relational graph convolution for spatial dependency modeling and temporal sequence modeling for dynamic pattern extraction.

The same problem formulation, multi-scale decomposition, and graph construction strategy are consistently applied to both the baseline and proposed models to ensure fair and rigorous comparative evaluation.

B. FRAMEWORK OVERVIEW (PROPOSED FRAMEWORK)

The proposed stock price prediction framework follows a modular and strictly controlled design to ensure fairness, reproducibility, and consistent architectural comparison with the REGCN baseline. An overview of the framework is illustrated in Fig. 1. As illustrated in Fig. 1, the proposed framework follows a unified multi-scale graph-based architecture in which all preprocessing and spatial modeling components are shared, while two alternative temporal modeling branches are introduced.

The pipeline begins with raw multivariate transaction data consisting of multiple financial attributes (e.g., open, close, RSI, SMA, volume). These attributes are first processed through a genetic algorithm (GA) optimized Variational Mode Decomposition (VMD) module, which decomposes each attribute into K intrinsic mode functions (IMFs). The GA module determines the optimal decomposition parameters (K, α) on the training set to ensure effective frequency separation while preventing information leakage. Each IMF subsequence is then processed independently through the same spatial modeling pipeline. For every decomposed component, relational graphs are constructed based on multiple dependency measures, including Pearson correlation, Spearman correlation, and Dynamic Time Warping. The resulting adjacency matrices are fused via a learnable weighted aggregation mechanism and symmetrically normalized before being fed into the multi-graph convolution module. The multi-graph convolution (Multi-GCN) block extracts relation-aware attribute embeddings at each time step. These embeddings serve as the shared spatial representation for both proposed

temporal modeling branches. From this point, the architecture bifurcates into two alternative temporal modules:

1) In the Refined REGCN branch, the graph-enhanced embeddings are processed by a simplified gated recurrent unit (Minimal GRU). The reset gate of the conventional GRU is removed to reduce parameter redundancy and improve stability. The recurrent updates capture sequential dependencies across the sliding window, and the final hidden state is used to generate IMF-level predictions.

2) In the GCN-ATTN branch, the graph-enhanced embeddings are processed by a dual multi-head self-attention module. This block computes scaled dot-product attention across temporal positions, enabling direct modeling of long-range dependencies. The attention output is passed through a position-wise feed-forward network (FFN) and subsequently through an adaptive gating mechanism. The adaptive gating module learns to balance original embeddings and attention-enhanced representations, allowing dynamic control over temporal information flow.

For each IMF component, the selected temporal branch produces a one-step-ahead prediction. Finally, IMF-level predictions are aggregated through summation to reconstruct the final closing price forecast, consistent with the additive property of VMD.

The overall pipeline consists of five main components: multi-scale sequence decomposition, relational graph construction, multi-graph convolution, temporal modeling, and prediction aggregation with unified evaluation.

a: Multi-Scale Sequence Decomposition.

Given a multivariate transaction time series

$$\mathbf{S}_T = (\mathbf{s}_0, \mathbf{s}_1, \dots, \mathbf{s}_{T-1}),$$

each transaction attribute is decomposed using Variational Mode Decomposition (VMD) into K intrinsic mode functions (IMFs). The decomposition can be expressed as

$$\mathbf{s}_t = \left[\sum_{k=1}^K u_t^{(1,k)}, \dots, \sum_{k=1}^K u_t^{(N,k)} \right]^\top. \quad (8)$$

where $u_t^{(n,k)}$ denotes the k -th IMF component of the n -th attribute. The decomposition parameters are optimized using a genetic algorithm on the training set only, and the same parameters are applied to the test set to prevent information leakage. Prediction is performed independently on each IMF, enabling multi-scale temporal modeling.

b: Relational Graph Construction.

For each decomposed subsequence, multiple relational graphs are constructed to model dependencies among transaction attributes. Let $\mathcal{G} = \{G_1, G_2, \dots, G_m\}$ denote the set of relational graphs, where each graph corresponds to a specific dependency measure, including Pearson correlation, Spearman's rank correlation, and Dynamic Time Warping (DTW). Each graph is represented by an adjacency matrix

$A_k \in \mathbb{R}^{N \times N}$, where N denotes the number of transaction attributes.

c: Multi-Graph Fusion and Convolution.

To integrate information from multiple relational graphs, a learnable weighted fusion mechanism is employed:

$$A_S = \sum_{k=1}^m w_k A_k, \quad w_k \in [0, 1], \quad (9)$$

where w_k are trainable scalar weights. The fused adjacency matrix is symmetrically normalized as

$$\hat{A}_S = D^{-\frac{1}{2}}(A_S + I)D^{-\frac{1}{2}}, \quad (10)$$

where D denotes the degree matrix and I is the identity matrix.

Graph convolution is then applied to extract spatial representations:

$$\mathbf{H}_t = \sigma(\hat{A}_S \mathbf{s}_t W), \quad (11)$$

where $W \in \mathbb{R}^{N \times d}$ is a learnable projection matrix, d denotes the hidden dimension, and $\sigma(\cdot)$ is a nonlinear activation function. This operation captures structural dependencies among transaction attributes at each time step.

d: Temporal Modeling.

The sequence of spatial embeddings $\{\mathbf{H}_t\}$ is subsequently processed by temporal modeling modules to capture dynamic dependencies. Unlike the original REGCN framework, which relies exclusively on gated recurrent units (GRUs), this study investigates two enhanced temporal modeling strategies:

- **Refined Recurrent Modeling:** a simplified and regularized recurrent architecture designed to improve stability and parameter efficiency while preserving sequential inductive bias.
- **Temporal Self-Attention Modeling:** a multi-head self-attention mechanism that captures long-range temporal dependencies in a fully parallel manner.

The general temporal prediction process can be expressed as

$$\hat{y}_{t+1}^{(k)} = f_\theta(\mathbf{H}_{t-L+1:t}), \quad (12)$$

where $f_\theta(\cdot)$ denotes the temporal modeling function parameterized by θ .

e: Prediction Aggregation and Evaluation.

Predictions from all decomposed subsequences are reconstructed via IMF-wise aggregation following Eq. (5).

All models share identical preprocessing, graph construction, and aggregation procedures. Therefore, any observed performance differences can be attributed solely to architectural innovations in the temporal modeling component.

C. SEQUENCE DECOMPOSITION (VMD + GA)

Financial time series are characterized by non-stationarity, noise, and multi-frequency fluctuation patterns, which increase forecasting complexity. To mitigate these effects and

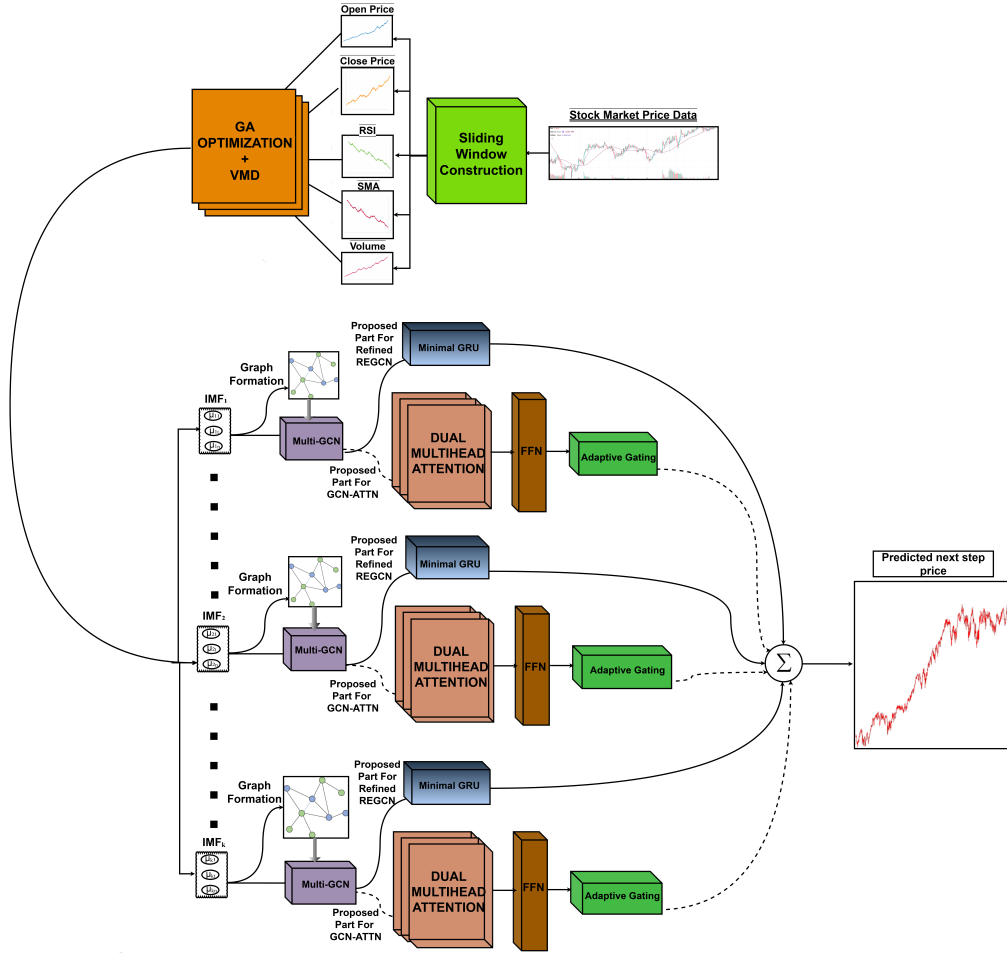


FIGURE 1. Proposed architecture of the multi-scale graph-based stock prediction framework(Refined REGCN + GCN-ATTN).

enhance model learning capacity, Variational Mode Decomposition (VMD) is employed to decompose each transaction attribute into multiple smoother sub-sequences with distinct frequency characteristics.

Given a univariate transaction attribute sequence $f(t)$, VMD decomposes it into K intrinsic mode functions (IMFs) $\{\mu_k(t)\}_{k=1}^K$ by solving the following constrained variational optimization problem:

$$\min_{\{\mu_k\}, \{\omega_k\}} \sum_{k=1}^K \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * \mu_k(t) \right] e^{-j\omega_k t} \right\|_2^2, \quad (13)$$

subject to the reconstruction constraint

$$\sum_{k=1}^K \mu_k(t) = f(t), \quad (14)$$

where $\delta(t)$ denotes the Dirac delta function, $*$ represents convolution, and ∂_t is the time derivative operator. Each IMF is assumed to be band-limited around a center frequency ω_k .

The quality of decomposition is primarily influenced by two hyperparameters: the number of IMFs K and the

quadratic penalty factor α . To automatically determine suitable values for (K, α) , a Genetic Algorithm (GA) is employed. The GA searches the parameter space through selection, crossover, and mutation operations.

The fitness function is defined based on the correlation between the original training signal and the residual component. For each transaction attribute, the residual sequence is computed as

$$\text{Res}(t) = f_{\text{train}}(t) - \sum_{k=1}^K \mu_k(t). \quad (15)$$

The fitness value is then defined as the average Pearson correlation coefficient between the original training sequence and the residuals:

$$\mathcal{F}(K, \alpha) = \frac{1}{N} \sum_{i=1}^N \rho_P(f_{\text{train}}^{(i)}(t), \text{Res}^{(i)}(t)), \quad (16)$$

where N denotes the number of transaction attributes and $\rho_P(\cdot)$ represents the Pearson correlation coefficient.

Minimizing $\mathcal{F}(K, \alpha)$ ensures that the residual component exhibits minimal linear correlation with the original signal, thereby improving decomposition effectiveness.

Importantly, GA optimization is performed exclusively on the training set. The optimized parameters (K, α) are subsequently applied unchanged to decompose the test set, preventing data leakage and ensuring experimental integrity.

After decomposition, each IMF subsequence is treated as an independent input sequence and processed by the graph-based prediction modules. The final stock price prediction is reconstructed by aggregating IMF-level forecasts as defined in Eq. (5), allowing the model to jointly capture low-frequency trends and high-frequency fluctuations.

D. GRAPH CONSTRUCTION (MULTI-RELATIONAL TRANSACTION GRAPHS)

To capture structural dependencies among transaction attributes, a set of multi-relational graphs is constructed for each stock. Unlike many graph-based stock prediction methods that model correlations across different stocks, the proposed framework focuses on modeling relationships among transaction attributes within a single stock. Such intra-attribute dependencies are structurally stable and provide meaningful inductive bias for price prediction.

Each transaction attribute is represented as a node in the graph. Formally, let $V = \{v_1, v_2, \dots, v_N\}$ denote the set of transaction attribute nodes, where N is the number of attributes. Multiple weighted undirected graphs are constructed to model different types of dependencies:

$$\mathcal{G} = \{G_1, G_2, \dots, G_m\}, \quad (17)$$

where each graph $G_k = (V, E_k)$ corresponds to a specific dependency measure.

Synchronous Correlation Graphs. To capture instantaneous dependencies among transaction attributes, Pearson correlation and Spearman's rank correlation coefficients are employed. Given two transaction attribute sequences $U = \{u_1, u_2, \dots, u_T\}$ and $V = \{v_1, v_2, \dots, v_T\}$, the Pearson correlation coefficient is defined as

$$\rho_P(U, V) = \frac{\sum_{t=1}^T (u_t - \mu_U)(v_t - \mu_V)}{\sqrt{\sum_{t=1}^T (u_t - \mu_U)^2} \sqrt{\sum_{t=1}^T (v_t - \mu_V)^2}}, \quad (18)$$

where μ_U and μ_V denote the mean values of sequences U and V , respectively.

Spearman's rank correlation coefficient captures monotonic relationships and is computed as the Pearson correlation between ranked sequences:

$$\rho_{Sp}(U, V) = 1 - \frac{6 \sum_{t=1}^T d_t^2}{T(T^2 - 1)}, \quad (19)$$

where d_t denotes the difference between the ranks of u_t and v_t .

Asynchronous Dependency Graph. To capture temporal misalignment and asynchronous similarity among transaction attributes, Dynamic Time Warping (DTW) is employed.

Given two sequences U and V , the DTW distance measures the minimum cumulative alignment cost between them. The distance is converted into a similarity measure through normalization:

$$\rho_{DTW}(U, V) = 1 - \frac{f_{DTW}(U, V)}{\kappa}, \quad (20)$$

where $f_{DTW}(\cdot)$ denotes the FastDTW distance and κ is a normalization constant used to scale values into a comparable range.

Adjacency Matrix Construction and Normalization.

For each relation type, an adjacency matrix $A_k \in \mathbb{R}^{N \times N}$ is constructed, where each element $a_{ij}^{(k)}$ represents the dependency strength between attributes i and j . To preserve self-information, identity matrices are added before normalization. Each adjacency matrix is then symmetrically normalized as

$$\hat{A}_k = D_k^{-\frac{1}{2}}(A_k + I)D_k^{-\frac{1}{2}}, \quad (21)$$

where D_k is the degree matrix of $A_k + I$ and I is the identity matrix.

The normalized adjacency matrices collectively represent complementary relational information. These matrices are subsequently fused within the graph convolution module using a learnable weighted aggregation mechanism. This multi-relational graph construction enables the model to comprehensively capture both synchronous and asynchronous dependencies among transaction attributes and provides the structural foundation for subsequent graph convolution and temporal modeling.

E. MULTI-GRAPH CONVOLUTION MODULE

To comprehensively model heterogeneous dependencies among transaction attributes, a multi-graph convolution module is adopted to integrate information from multiple relational graphs. Each relational graph captures a distinct type of dependency, and their joint utilization enables richer and more robust structural representation learning.

Let $\{\hat{A}_1, \hat{A}_2, \dots, \hat{A}_m\}$ denote the pre-normalized adjacency matrices obtained from the graph construction stage.

Since different relational graphs provide complementary structural information, a learnable weighted fusion mechanism is employed to combine them:

$$A_S = \sum_{k=1}^m w_k \hat{A}_k, \quad (22)$$

where $w_k \in [0, 1]$ are trainable non-negative parameters. This adaptive fusion strategy allows the model to automatically adjust the contribution of each relation type during training.

Given the transaction attribute feature vector $\mathbf{s}_t \in \mathbb{R}^N$ at time step t , graph convolution is performed as

$$\mathbf{h}_t = \sigma(W^\top A_S \mathbf{s}_t), \quad (23)$$

where $W \in \mathbb{R}^{N \times d}$ is a trainable weight matrix, d denotes the embedding dimension, and $\sigma(\cdot)$ is a nonlinear activation function.

The resulting embedding $\mathbf{h}_t$ represents a relation-aware transformation of transaction attributes that integrates both feature information and multi-relational topology.

By applying this operation at each time step, the original transaction sequence is transformed into a graph-enhanced representation sequence:

$$\mathbf{H} = (\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_T). \quad (24)$$

Importantly, the multi-graph convolution module is shared across the baseline REGCN model and both proposed architectures. This controlled design ensures that any observed performance improvements can be attributed exclusively to innovations in temporal modeling rather than differences in spatial dependency modeling.

F. TEMPORAL MODELING AND PROPOSED ARCHITECTURES

After obtaining relation-aware embeddings through the multi-graph convolution module, temporal dependencies across historical observations must be modeled to generate accurate stock price predictions. While the baseline REGCN employs gated recurrent units (GRUs), this study proposes two alternative architectures under the same preprocessing and graph modeling framework.

1) GCN with Temporal Self-Attention

The first proposed model replaces recurrent temporal modeling with a temporal self-attention mechanism. Recurrent architectures process sequences sequentially, which may limit their ability to capture long-range dependencies and restrict parallelization. In contrast, temporal self-attention enables direct interaction between all time steps within the input window.

Given an input window of length L , let

$$\mathbf{H} = (\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_L) \quad (25)$$

denote the sequence of graph-enhanced embeddings, where

$$\mathbf{h}_t = \sigma(W_{\text{gcn}}^\top A_S \mathbf{s}_t), \quad (26)$$

with $A_S = \sum_{k=1}^m w_k \hat{A}_k$ representing the fused normalized adjacency matrix.

Temporal self-attention computes:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V, \quad (27)$$

where Q, K , and V are linear projections of $\mathbf{H}$ and d denotes the embedding dimension.

A multi-head attention mechanism is employed to capture temporal dependencies from multiple representation subspaces. The attention output is further processed by a position-wise feed-forward network, followed by residual connections and layer normalization.

To adaptively balance original embeddings and attention-enhanced representations, a gating mechanism is introduced:

$$\mathbf{H}' = G \odot \mathbf{H}_{\text{attn}} + (1 - G) \odot \mathbf{H}, \quad (28)$$

where $G = \sigma(W_g \mathbf{H}_{\text{attn}})$ is a learnable gate.

Finally, a time-distributed linear layer generates step-wise predictions, and the final closing price prediction is obtained from the last time step of the output sequence.

This architecture improves parallelism and enhances long-range temporal dependency modeling while remaining fully compatible with the multi-graph convolution module.

2) Refined REGCN Architecture

The second proposed model introduces a refined recurrent architecture that preserves the sequential modeling philosophy of REGCN while improving parameter efficiency and training stability.

First, the fused normalized adjacency matrix is computed as

$$A_S = \sum_{k=1}^m w_k \hat{A}_k, \quad (29)$$

where $\hat{A}_k = D_k^{-1/2}(A_k + I)D_k^{-1/2}$ and w_k are learnable fusion weights.

At each time step, graph convolution is applied:

$$\mathbf{x}_t = \sigma(W_{\text{gcn}}^\top A_S \mathbf{s}_t), \quad (30)$$

which serves as the input to a simplified gated recurrent update.

Unlike the standard GRU, the refined architecture removes the reset gate to reduce parameter redundancy and stabilize optimization. The update equations are defined as

$$\mathbf{z}_t = \sigma(W_z \mathbf{x}_t + U_z \mathbf{h}_{t-1} + \mathbf{b}_z), \quad (31)$$

$$\tilde{\mathbf{h}}_t = \tanh(W_h \mathbf{x}_t + U_h \mathbf{h}_{t-1} + \mathbf{b}_h), \quad (32)$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t. \quad (33)$$

This simplified gating structure reduces parameter count while retaining the ability to model temporal dependencies.

Furthermore, normalized adjacency matrices are cached and reused across time steps to reduce computational overhead. Gradient clipping is applied during optimization to enhance numerical stability and prevent gradient explosion.

The refined REGCN maintains interpretability and sequential dependency modeling while offering improved robustness and efficiency. Together with the temporal self-attention architecture, these two proposed models provide complementary strategies for enhancing temporal modeling within a unified graph-based framework.

G. LOSS FUNCTION AND TRAINING STRATEGY

In stock price prediction, both numerical accuracy and directional consistency are crucial for practical financial decision-making. Numerical accuracy evaluates the deviation between predicted and actual prices, while directional consistency measures whether the model correctly captures upward or downward price movement. To balance these two objectives, a trend-aware composite loss function is adopted.

Trend-Aware Training Loss. The overall loss function is defined as

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + r_{\text{ACC}} \cdot \mathcal{L}_{\text{trend}}, \quad (34)$$

where r_{ACC} is a fixed weighting coefficient controlling the contribution of the trend regularization term.

Following the original REGCN baseline study, the coefficient is set as:

$$r_{ACC} = 0.1.$$

These values are directly inherited from the baseline to ensure fair comparison and experimental consistency across all models.

The mean squared error term is defined as

$$\mathcal{L}_{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad (35)$$

which enforces numerical precision of price predictions.

The trend regularization term penalizes incorrect directional predictions relative to the previous observed price:

$$\mathcal{L}_{trend} = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(\text{sign}(y_i - y_{i-1}) \neq \text{sign}(\hat{y}_i - \hat{y}_{i-1})), \quad (36)$$

where $\mathbb{I}(\cdot)$ denotes the indicator function.

Importantly, the loss is computed only on the final prediction of each input window. Given a sliding window of length L , only the prediction at time step L contributes to the optimization objective. This formulation ensures consistency across recurrent and attention-based architectures.

Training Strategy. All models are trained using the Adam optimizer with a fixed learning rate under mini-batch gradient descent. Sliding windows are extracted from historical sequences to construct training samples.

During training, the previous closing price from the last time step of each input window is explicitly used to compute the trend regularization term, ensuring that no future information is utilized.

For the refined REGCN architecture, gradient clipping is applied during optimization to mitigate gradient explosion in recurrent updates. Additionally, normalized adjacency matrices are precomputed and cached for each stock to reduce computational overhead without altering model behavior.

Trend Evaluation Protocol. During evaluation, directional prediction accuracy is measured differently for the baseline and proposed models.

For the baseline REGCN model, trend accuracy is computed using the standard sign-based criterion:

$$\text{Trend}(y_t) = \text{sign}(y_t - y_{t-1}). \quad (37)$$

The baseline results are reported using its original evaluation protocol to maintain consistency with the published REGCN framework and ensure fair comparison.

For the proposed GCN-ATTN and refined REGCN models, a margin-based trend criterion is employed to avoid penalizing insignificant price fluctuations. A dynamic threshold ε is defined as:

$$\varepsilon = \gamma \cdot \mathbb{E}[|y_t - y_{t-1}|], \quad (38)$$

where γ is a fixed scaling constant and the expectation is computed over the test set.

Under this criterion, a price movement is considered significant only if

$$|y_t - y_{t-1}| > \varepsilon.$$

Directional correctness is evaluated only for movements exceeding this threshold.

This margin-based evaluation is applied exclusively during testing and does not influence the training objective.

By maintaining identical loss coefficients across all models and introducing margin-based evaluation only for the proposed architectures, the experimental setup ensures both fairness to the baseline and a more realistic assessment of directional prediction performance.

H. PREDICTION AGGREGATION AND EVALUATION PROTOCOL

IMF-wise Prediction Aggregation. After applying Variational Mode Decomposition (VMD), each original stock price sequence is decomposed into K intrinsic mode functions (IMFs). Each IMF subsequence is modeled independently using the same graph construction and temporal modeling framework. The final prediction sequence is reconstructed via IMF-wise aggregation as defined in Eq. (5).

This aggregation preserves the additive property of VMD and allows multi-scale temporal information to contribute to the final output.

Temporal Alignment Before Evaluation. Due to sliding-window prediction and IMF-wise modeling, predicted sequences and ground-truth sequences may exhibit slight length mismatches. To ensure consistent and fair evaluation, all sequences are aligned before computing performance metrics.

Let $\hat{Y}$ denote the predicted sequence, Y the ground-truth sequence, and Y_{prev} the previous-price reference sequence. The aligned evaluation length is defined as

$$T = \min(|\hat{Y}|, |Y|, |Y_{\text{prev}}|). \quad (39)$$

All sequences are truncated to the first T samples prior to metric computation. This alignment procedure prevents indexing inconsistencies and ensures comparability across models.

Evaluation Metrics. Model performance is evaluated using both numerical accuracy and directional consistency metrics.

Numerical prediction quality is measured using:

- Root Mean Squared Error (RMSE),
- Mean Absolute Error (MAE),
- Coefficient of determination (R^2).

Trend accuracy is computed by comparing predicted and actual price movement directions relative to the previous observed price.

For the baseline REGCN model, directional accuracy is computed using the standard sign-based criterion:

$$\text{Trend}(y_t) = \text{sign}(y_t - y_{t-1}). \quad (40)$$

For the proposed GCN-ATTN and refined REGCN models, a margin-based criterion is employed to avoid penalizing in-

significant price fluctuations. A dynamic margin ε is defined as

$$\varepsilon = \gamma \cdot \mathbb{E} [|y_t - y_{t-1}|], \quad (41)$$

where γ is a fixed scaling constant and the expectation is computed over the test set.

Only price movements satisfying

$$|y_t - y_{t-1}| > \varepsilon$$

are considered meaningful trends during evaluation.

Aggregated Evaluation Across Stocks. In addition to per-stock evaluation, aggregated performance metrics are reported across all stocks to assess overall predictive capability.

Let $\hat{Y}^{(i)}$ and $Y^{(i)}$ denote the predicted and true sequences for the i -th stock. Aggregated evaluation is performed by concatenating predictions and ground-truth sequences across all stocks:

$$\hat{Y}_{\text{all}} = \text{concat}_i (\hat{Y}^{(i)}), \quad Y_{\text{all}} = \text{concat}_i (Y^{(i)}). \quad (42)$$

Performance metrics are then computed on the concatenated sequences. This aggregated protocol follows the original REGCN experimental setting and provides a comprehensive assessment of model generalization across multiple stocks.

By employing IMF-wise reconstruction, strict temporal alignment, and consistent evaluation criteria, the proposed framework ensures that performance improvements are attributable to architectural innovations rather than evaluation artifacts.

IV. EXPERIMENTAL SETUP

This section describes the datasets, feature construction, baseline models, evaluation metrics, and implementation details used to evaluate the proposed models. To ensure fairness and reproducibility, all experiments strictly follow the preprocessing, training strategy, and evaluation protocol of the original REGCN framework [18]. The same experimental pipeline is applied to all baseline and proposed models.

A. DATASETS

To evaluate the proposed models under diverse market structures and trading frequencies, experiments are conducted on three representative datasets: the Shanghai Stock Exchange (SSE), the Dow Jones Industrial Average (DJIA), and a newly constructed NASDAQ dataset. These datasets collectively cover emerging and developed markets, as well as daily and high-frequency trading settings, enabling a comprehensive empirical evaluation.

1) SSE Dataset

The Shanghai Stock Exchange (SSE) dataset consists of daily transaction data for stocks included in the SSE 50 index, representing large-cap and highly liquid equities in the Chinese stock market. The dataset spans from January 2021 to December 2023, covering multiple market regimes with varying volatility conditions.

Following the experimental setting of REGCN [18], the original study utilized 40 SSE stocks. However, in our implementation, 24 SSE stocks are used. This adjustment was necessary because part of the publicly available stock data provided with the original framework contained corrupted or incomplete records that could not be reliably processed. To ensure data integrity and consistent preprocessing across all experiments, only stocks with complete and valid transaction records were retained. The list of selected SSE stock codes is provided in Table 1.

Each stock in the SSE dataset is characterized by the following transaction-related features:

- Opening price
- Closing price
- Highest price
- Lowest price
- Adjusted closing price
- Absolute price change
- Percentage price change
- Trading volume
- Turnover value

These features jointly capture price evolution, trading intensity, and liquidity characteristics, making the dataset suitable for modeling complex temporal dynamics in emerging markets.

2) DJIA Dataset

The Dow Jones Industrial Average (DJIA) dataset contains daily transaction data for 30 U.S. blue-chip stocks listed on major exchanges. These companies represent industry-leading corporations with substantial market capitalization and relatively stable trading activity. The dataset covers the same time period as the SSE dataset to facilitate consistent cross-market comparison.

For each DJIA stock, the following core transaction features are included:

- Opening price
- Closing price
- Highest price
- Lowest price
- Adjusted closing price
- Trading volume

Compared to the SSE dataset, DJIA stocks generally exhibit lower volatility and stronger cross-sectional correlations, which makes this dataset suitable for assessing graph-based relational modeling. The full list of DJIA stock symbols is also shown in Table 1.

3) NASDAQ Dataset

To further evaluate model generalization under higher-frequency trading conditions, a NASDAQ dataset is constructed using the Alpha Vantage API. This dataset contains hourly transaction data for nine NASDAQ-listed companies: ADBE, BABA, CRM, MS, ORCL, PYPL, QCOM, V, and

TABLE 1. Stock codes included in the datasets

Dataset	Stock Codes
SSE	600000, 600015, 600016, 600018, 600028, 600036, 600048, 600050, 600104, 600111, 600585, 600637, 600795, 600837, 600887, 601006, 601088, 601166, 601169, 601186, 601211, 601288, 601318, 601328
DJIA	AAPL, AMGN, BA, CAT, CRM, CSCO, CVX, DIS, GS, HD, HON, IBM, INTC, JNJ, JPM, KO, MCD, MMM, MRK, MSFT, NKE, PG, TRV, UNH, V, VZ, WBA, WMT
NASDAQ	ADBE, BABA, CRM, MS, ORCL, PYPL, QCOM, V, WFC

WFC, spanning multiple industry sectors including technology, financial services, and consumer payments.

Each stock contains 3,744 hourly observations from January 2023 to June 30, 2025, corresponding to six trading hours per business day. In addition to standard OHLCV variables, a set of technical and market-related indicators is incorporated to enrich the feature space.

The NASDAQ feature set includes:

- Open, High, Low, and Close prices
- Trading volume
- Trend strength indicators: ADX_{14} , $ADXR_{14}$, DI^+ , and DI^-
- Momentum indicators: RSI_{14} and MACD histogram
- Volatility indicator: ATR_{14}
- Moving averages: SMA_5 and SMA_{20}
- Bollinger Bands: upper, middle, and lower bands
- Market co-movement feature: rolling correlation with the SPY ETF

The NASDAQ stock symbols are listed in Table 1. By combining price-based, volume-based, technical, and market-relation features, this dataset provides a higher-dimensional and higher-frequency evaluation setting.

Across all datasets, normalization parameters are computed exclusively from the training set to prevent information leakage. A sliding window strategy is employed to construct supervised learning samples, where a fixed-length historical sequence is used to predict the next-step closing price. The same preprocessing procedure is applied uniformly to all datasets and models to ensure fair comparison.

B. DATA PREPROCESSING

To ensure numerical stability and facilitate efficient model training, all input features are normalized using Min–Max scaling, which rescales each feature to the interval $[0, 1]$:

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}. \quad (43)$$

To strictly prevent information leakage, the normalization parameters $\min(x)$ and $\max(x)$ are computed exclusively

from the training subset of each stock time series. The same parameters are subsequently applied to the corresponding validation and test subsets. After model inference, inverse normalization is performed to restore predicted values to their original price scale prior to metric computation.

Following the experimental protocol of REGCN [18], a sliding window strategy is adopted to transform each stock time series into supervised learning samples. Given a historical observation sequence of fixed length L , the model predicts the closing price at the next time step (one-step-ahead forecasting). In the main experiments, a fixed window length is used for model comparison, while additional window lengths ($L = 15, 30, 45$) are evaluated in the ablation study to analyze sensitivity to temporal context size.

All datasets are split chronologically into training, validation, and test subsets, preserving temporal order to reflect realistic forecasting conditions and avoid look-ahead bias. The same preprocessing pipeline, window construction procedure, and normalization strategy are uniformly applied across all baseline and proposed models to ensure a fair and consistent comparison.

C. VARIATIONAL MODE DECOMPOSITION PARAMETERS

Following the REGCN framework [18], Variational Mode Decomposition (VMD) is applied independently to each stock time series to extract intrinsic mode functions (IMFs) corresponding to different temporal scales. VMD decomposes non-stationary financial signals into a finite number of band-limited components, enabling effective separation of long-term trends and short-term market fluctuations prior to temporal modeling.

The decomposition process is governed by two key hyperparameters: the number of modes K and the quadratic penalty factor α , which controls the bandwidth constraint of each mode. Rather than adopting a uniform configuration across all stocks, the pair (K, α) is optimized separately for each stock to account for heterogeneous volatility structures, liquidity profiles, and trading characteristics observed across different markets.

Consistent with the methodology described in Section III, the optimal VMD parameters are determined using a Genetic Algorithm (GA) applied exclusively to the training subset of each stock. The optimized parameters are then fixed and applied unchanged to the validation and test sets to prevent information leakage. This stock-wise tuning strategy allows the decomposition process to adapt to asset-specific dynamics while maintaining strict experimental integrity.

The resulting optimal values of K and α for all stocks across the SSE, DJIA, and NASDAQ datasets are summarized in Table 2. As shown in the table, the number of modes typically ranges between 2 and 4, reflecting moderate multi-scale complexity in the examined financial series. The penalty factor α varies across stocks, indicating differences in bandwidth regularization requirements under distinct market conditions.

By performing per-stock parameter optimization and maintaining consistent train–validation–test separation, the VMD

preprocessing stage enhances signal decomposition quality without introducing evaluation bias.

D. BASELINE MODELS

To assess the effectiveness of the proposed architectures, their performance is compared against a diverse set of baseline models spanning statistical, deep learning, decomposition-based, and graph-based forecasting approaches. These baselines represent widely adopted methodologies in financial time-series prediction and reflect different assumptions regarding temporal dynamics, nonlinearity, and relational modeling.

Statistical Models. Classical statistical forecasting methods are included as foundational benchmarks. The Autoregressive Integrated Moving Average (ARIMA) model is employed to capture linear temporal dependencies through autoregressive and moving-average components [1]. Although ARIMA assumes linearity and stationarity, it remains a commonly used reference model in financial forecasting studies. In addition, the Component Generalized Autoregressive Conditional Heteroskedasticity (CGARCH) model is adopted to explicitly model volatility clustering and conditional variance dynamics, which are prominent features of financial time series [3]. These models provide interpretable reference points for evaluating the benefits of nonlinear and graph-based approaches.

Deep Learning Models. To model nonlinear temporal dependencies, several neural network architectures are considered. Long Short-Term Memory (LSTM) networks are included due to their ability to capture long-range sequential dependencies through gated memory mechanisms [5]. Gated Recurrent Units (GRUs) are also evaluated as a computationally efficient alternative with reduced gating complexity [6]. In addition, a hybrid CNN–LSTM model is implemented, where convolutional layers first extract local temporal patterns before recurrent modeling [4]. These architectures represent mainstream deep learning approaches that focus primarily on temporal modeling without explicitly incorporating relational graph structures.

Decomposition-Based Models. To address the multi-scale and non-stationary nature of financial time series, decomposition-driven forecasting models are incorporated. Specifically, the Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) framework is combined with deep learning predictors to decompose raw price series into intrinsic mode functions representing distinct frequency components [15]. These methods aim to improve robustness by modeling decomposed components independently.

Graph-Based Models. Graph neural network-based baselines are included to explicitly capture structured dependencies. The Temporal Graph Convolutional Network (TG-CN) integrates graph convolution with recurrent modeling to jointly learn spatial and temporal correlations [11]. Furthermore, the Relational Embedded Graph Convolutional Network (REGCN) [18] is employed as the primary refer-

ence model. REGCN combines VMD-based decomposition, multi-graph convolution, and recurrent temporal modeling, and serves as the most directly comparable baseline to the proposed architectures.

Fair Comparison Protocol. To ensure consistency, all neural network-based models are trained using the same preprocessing pipeline, sliding window construction, batch size (128), number of training epochs (100), and learning rate configuration. Chronological train–validation–test splits are maintained across all datasets. For the proposed models, random seed initialization is fixed (seed = 42) to enhance reproducibility.

Evaluation metrics and aggregation procedures follow the unified protocol described in Section IV-E. The baseline REGCN model is evaluated using its original trend criterion, while the proposed models additionally report margin-based directional accuracy as described earlier. This controlled experimental setup ensures that observed performance differences primarily reflect architectural design choices rather than variations in training or evaluation settings.

E. EVALUATION METRICS

The predictive performance of all models is evaluated using a combination of regression-based accuracy metrics and directional trend metrics. This dual evaluation framework enables assessment of both numerical precision and practical directional forecasting capability, which is particularly relevant in financial decision-making contexts.

Trend Accuracy (ACC). Directional accuracy evaluates whether the predicted price movement direction matches the actual direction between consecutive time steps. Let n_{correct} denote the number of instances where the predicted and actual price differences share the same sign, and n denote the total number of evaluated samples. Trend accuracy is computed as:

$$\text{ACC} = \frac{n_{\text{correct}}}{n}. \quad (44)$$

For the baseline REGCN model, directional accuracy is computed using the standard sign-based criterion, consistent with the original study [18].

For the proposed models, a margin-based trend criterion is additionally applied during evaluation. Specifically, small price fluctuations below a predefined dynamic threshold are excluded from directional counting in order to reduce noise-induced misclassification. This margin-based evaluation provides a more stable assessment of meaningful trend movements while preserving comparability with the baseline results.

Coefficient of Determination (R^2). The coefficient of determination measures the proportion of variance in the true stock prices that is explained by the model predictions. It is defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^n (X_i - Y_i)^2}{\sum_{i=1}^n (X_i - \bar{X})^2}, \quad (45)$$

where X_i and Y_i denote the true and predicted prices, respectively, and $\bar{X}$ represents the mean of the ground-truth values. Higher R^2 values indicate stronger explanatory capability.

TABLE 2. Optimal VMD parameters in datasets

Dataset	Stock	K	α	Stock	K	α	Stock	K	α
SSE	600000	3	296	600104	4	339	601088	4	311
	600015	3	331	600111	4	402	601166	4	293
	600016	3	331	600585	3	367	601169	4	308
	600018	4	371	600637	3	314	601186	3	344
	600028	4	336	600795	4	575	601211	3	299
	600036	4	404	600837	4	376	601288	4	302
	600048	4	373	600887	3	315	601318	3	330
	600050	3	305	601006	4	298	601328	3	307
DJIA	AAPL	3	417	HON	3	422	NKE	3	357
	AMGN	4	415	IBM	4	304	PG	4	366
	BA	4	389	INTC	4	346	TRV	4	317
	CAT	4	342	JNJ	4	324	UNH	4	305
	CRM	3	339	JPM	2	337	V	4	386
	CSCO	3	327	KO	2	331	VZ	4	353
	CVX	4	602	MCD	4	306	WBA	3	334
	DIS	3	345	MMM	4	372	WMT	4	324
	GS	4	348	MRK	4	310			
	HD	4	334	MSFT	3	314			
NASDAQ	ADBE	4	298	BABA	4	302	CRM	4	274
	MS	4	357	ORCL	4	216	PYPL	4	229
	QCOM	4	211	V	4	218	WFC	4	210

Root Mean Squared Error (RMSE). RMSE quantifies the square root of the mean squared prediction error:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (X_i - Y_i)^2}. \quad (46)$$

Due to its quadratic form, RMSE penalizes larger prediction deviations more heavily and is sensitive to extreme forecasting errors.

Mean Absolute Error (MAE). MAE measures the average absolute deviation between predicted and actual prices:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |X_i - Y_i|. \quad (47)$$

Compared to RMSE, MAE provides a more interpretable measure of average error magnitude and is less influenced by outliers.

Evaluation Protocol. All regression metrics are computed on inverse-normalized predictions to ensure evaluation in the original price scale. Following the REGCN protocol [18], predictions are first aligned temporally and then aggregated across all stocks before metric computation. This aggregated evaluation strategy provides a comprehensive assessment of model generalization across different equities and market conditions.

F. IMPLEMENTATION DETAILS

All experiments are implemented in TensorFlow and executed on an NVIDIA Tesla P100 GPU platform. The Adam optimizer is employed for all models with a fixed learning rate of 8×10^{-4} . To maintain a controlled experimental environment,

a consistent batch size of 128 and a fixed training duration of 100 epochs are used across all baseline and proposed models. Early stopping is not applied; instead, each model is trained for the full number of epochs under identical conditions.

To analyze the effect of temporal context length, three sliding window sizes ($L = 15, 30$, and 45) are evaluated on all datasets. These window lengths are applied uniformly to both baseline and proposed models, and the corresponding results are discussed in the Results and discussion section.

For reproducibility, experiments involving the proposed models are conducted using a fixed random seed of 42. Baseline implementations follow the configurations recommended in their respective original studies, with consistent data pre-processing and training schedules.

To improve numerical stability during optimization, gradient clipping is applied to both proposed architectures. This stabilization is particularly important for recurrent updates and attention-based temporal modeling when processing longer sequences. Baseline models are trained using their standard optimization procedures without additional stabilization techniques.

The attention-based temporal model incorporates dropout with a rate of 0.1 within the self-attention module to mitigate potential overfitting. Dropout is not applied to the refined recurrent architecture or to statistical baselines. No weight decay regularization is used in any of the evaluated models.

In addition to predictive accuracy, computational efficiency is assessed by recording training time per epoch for all models under identical hardware conditions. Inference time is also measured to provide a transparent comparison of runtime scalability.

All experiments adhere to the same preprocessing, graph construction, VMD decomposition, and evaluation protocols described in the REGCN framework [18], ensuring that observed performance differences arise from architectural modifications rather than inconsistencies in experimental setup.

V. RESULTS AND DISCUSSION

A. EXPERIMENTAL SETUP AND EVALUATION PROTOCOL

This subsection describes the experimental configuration and evaluation protocol adopted to assess the effectiveness of the proposed models. All experiments are conducted under a unified and controlled setting to ensure fair and reproducible comparisons with baseline methods.

Datasets and Experimental Settings. Experiments are performed on three stock market datasets described in Section 3. For each dataset, an identical preprocessing pipeline is applied, including Min–Max normalization, sequence decomposition using Variational Mode Decomposition (VMD), sliding-window sample construction, and multi-relational transaction graph construction. An 80%/20% chronological split is used to divide the data into training and testing subsets. The same data partitions are preserved across all models to prevent information leakage and to ensure that performance differences are attributable to architectural design rather than data inconsistencies.

Sequence Length Configuration. To evaluate robustness with respect to historical context length, three input sequence lengths ($L = 15, 30, 45$) are considered. For each sequence length, baseline and proposed models are trained independently while keeping preprocessing, graph construction, optimizer settings, and training epochs unchanged. This design enables a controlled ablation analysis of the impact of temporal context on forecasting performance, as discussed in Section 4.3.

Baseline and Comparison Models. The proposed architectures are compared against representative statistical models (ARIMA, CGARCH), deep learning sequence models (LSTM, GRU, CNN-LSTM), graph-based models (T-GCN, REGCN), and decomposition-based approaches (CEEMDAN-based models). All baseline implementations follow configurations reported in their respective studies or standard implementations in the literature. Importantly, all models operate on the same preprocessed inputs and identical train–test splits to ensure comparability.

Evaluation Metrics. Model performance is evaluated using both numerical accuracy metrics and directional trend measures. Numerical prediction accuracy is assessed using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2). Directional forecasting capability is measured using trend accuracy (ACC), which evaluates consistency between predicted and actual price movement directions relative to the previous observed price.

For the baseline REGCN model, trend accuracy is computed using the standard sign-based criterion, consistent with its original evaluation protocol. For the proposed models

(GCN-ATTN and Refined REGCN), a margin-based trend evaluation strategy is applied during testing to avoid penalizing negligible price fluctuations, thereby providing a more stable assessment of directional performance. This margin-based strategy is used exclusively during evaluation and does not modify the training objective.

Runtime Measurement. To assess computational efficiency, training time and inference time are recorded for all models under identical hardware and software environments. These runtime statistics are reported alongside predictive metrics to analyze the trade-off between forecasting accuracy and computational cost.

Aggregated Evaluation Protocol. In addition to per-stock evaluation, aggregated metrics are computed by concatenating predictions and corresponding ground-truth values across all stocks within each dataset. This protocol follows the evaluation strategy adopted in the original REGCN framework and provides a comprehensive assessment of generalization across multiple securities.

By maintaining consistent preprocessing procedures, graph construction strategies, hyperparameter settings, and evaluation criteria across all models, the reported results primarily reflect differences in model architecture rather than experimental variations.

B. OVERALL PERFORMANCE COMPARISON ACROSS MODELS

This subsection presents a comprehensive quantitative comparison between the proposed models and representative baseline approaches. The evaluation examines numerical prediction accuracy, directional consistency, and computational efficiency under the unified experimental setting described in Section 4.1.

All models are trained and evaluated using identical preprocessing procedures, VMD configurations, relational graph construction strategies, and chronological train–test splits. Results are reported separately for each dataset using a fixed input sequence length of 30, which was identified in Section 4.3 as providing an effective balance between contextual richness and noise suppression. All metrics are computed under the aggregated evaluation protocol to assess overall generalization across multiple stocks.

Comparison Models. The evaluated methods span multiple modeling paradigms, including classical statistical models (ARIMA, CGARCH), deep learning sequence architectures (LSTM, GRU, CNN-LSTM), graph-based models (T-GCN, REGCN), and hybrid decomposition-based approaches (CEEMDAN-BiLSTM). These baselines represent diverse methodological categories and serve as appropriate reference points for assessing the effectiveness of the proposed architectures. The two proposed models are: (i) GCN-ATTN, which replaces recurrent temporal modeling with multi-head self-attention, and (ii) Refined REGCN, which introduces architectural simplifications and training stabilizations while preserving the structural philosophy of REGCN.

TABLE 3. Per-stock performance on SSE dataset (Sequence Length = 30)

Stock	REGCN						GCN-ATTN						Refined REGCN					
	ACC	R^2	RMSE	MAE	Train	Test	ACC	R^2	RMSE	MAE	Train	Test	ACC	R^2	RMSE	MAE	Train	Test
600000	0.4483	0.7700	0.1250	0.1012	515.64	41.8732	0.5345	0.9062	0.0799	0.0694	254.52	19.8426	0.5086	0.8126	0.1129	0.0953	69.49	3.1847
600015	0.7414	0.8625	0.0350	0.0287	513.38	42.5561	0.7259	0.7925	0.0673	0.0542	239.74	20.4187	0.8276	0.8929	0.0309	0.0241	65.31	3.2965
600016	0.6983	0.8388	0.0392	0.0288	685.11	43.1024	0.5603	0.9066	0.1594	0.1565	322.64	21.0365	0.7241	0.8487	0.0380	0.0288	86.18	3.3428
600018	0.6121	0.9396	0.0350	0.0280	689.31	40.9987	0.6207	0.8766	0.0500	0.0435	332.71	19.7743	0.6552	0.9404	0.0348	0.0289	89.71	3.1189
600028	0.5862	0.9562	0.0827	0.0677	679.91	41.6645	0.6172	0.9025	0.2154	0.1902	332.42	20.1129	0.6293	0.9462	0.0916	0.0724	92.31	3.2674
600036	0.5000	0.6261	1.3134	1.0389	510.57	42.3189	0.6207	0.9014	0.6745	0.4938	245.60	20.6874	0.5345	0.9014	0.6746	0.5168	65.49	3.3016
600048	0.4741	0.9577	0.3279	0.2542	683.39	43.4516	0.5259	0.9624	0.3092	0.2573	330.35	21.1948	0.5345	0.9537	0.3433	0.2607	88.19	3.3892
600050	0.7672	0.9721	0.0638	0.0481	684.56	40.7423	0.7914	0.8855	0.1293	0.1059	327.01	19.5632	0.8276	0.9791	0.0552	0.0439	87.40	3.1453
600104	0.6983	0.7427	0.2729	0.2253	684.32	42.9075	0.6034	0.7544	0.7128	0.6478	328.07	20.9536	0.7500	0.7245	0.2825	0.2288	88.82	3.2588
600111	0.6121	0.9311	0.4757	0.3894	685.42	41.2358	0.7241	0.9730	0.2977	0.2227	326.10	19.9875	0.6552	0.9263	0.4918	0.4001	86.52	3.2146
600585	0.4741	0.6674	0.9265	0.8758	681.61	42.6894	0.7845	0.9459	0.3735	0.2779	324.64	20.5413	0.4052	0.5373	1.0928	1.0225	90.53	3.3341
600637	0.6034	0.9283	0.0999	0.0774	513.50	43.0082	0.6828	0.9377	0.1503	0.1142	246.35	21.0827	0.5517	0.9158	0.1083	0.0853	67.29	3.3725
600795	0.6638	0.9457	0.0494	0.0406	692.17	40.5569	0.6293	0.9189	0.0603	0.0507	325.12	19.4328	0.6897	0.9423	0.0509	0.0423	89.81	3.1097
600837	0.6293	0.8208	0.1547	0.1172	515.63	41.9984	0.6034	0.8614	0.1785	0.1414	243.50	20.2649	0.6207	0.8318	0.1499	0.1190	68.69	3.2419
600887	0.4397	-2.3004	1.4066	1.3160	682.53	42.7713	0.4397	-3.2617	1.5984	1.5384	328.19	20.8731	0.4052	-0.3880	0.9122	0.8383	90.04	3.2863
601006	0.5000	0.0106	0.0853	0.0613	691.31	43.2245	0.5345	0.1180	0.0806	0.0615	322.35	21.2684	0.5862	0.1999	0.0767	0.0547	90.73	3.3584
601088	0.5690	0.8341	0.3661	0.2989	689.48	40.8896	0.6207	0.8485	0.5552	0.4817	311.14	19.6589	0.6638	0.9369	0.3584	0.3046	90.23	3.1726
601166	0.4052	-0.1296	0.7791	0.6454	708.56	41.4472	0.4397	0.8874	0.2460	0.1937	309.47	20.3256	0.3879	0.1867	0.6610	0.5353	92.23	3.2298
601169	0.5776	0.3609	0.0514	0.0385	690.40	42.5348	0.6086	0.4008	0.0498	0.0410	340.65	20.7098	0.5172	0.2796	0.0546	0.0396	89.41	3.3094
601186	0.5862	0.9671	0.1902	0.1311	515.94	43.6671	0.6172	0.9483	0.2383	0.1716	245.04	21.1462	0.6034	0.9751	0.1655	0.1151	67.06	3.3476
601211	0.6034	0.8292	0.1538	0.1189	519.59	40.3347	0.6983	0.8613	0.1386	0.0999	243.80	19.8047	0.6810	0.8486	0.1448	0.1064	67.53	3.1362
601288	0.5086	0.4805	0.0599	0.0519	687.18	41.7865	0.4828	0.5395	0.0676	0.0581	321.09	20.4975	0.4741	-0.0976	0.0871	0.0801	91.28	3.2641
601318	0.5948	0.9608	0.8063	0.6373	515.44	42.1193	0.7069	0.9788	0.5924	0.4721	246.38	20.1583	0.5948	0.9575	0.8390	0.6596	67.03	3.2957
601328	0.5690	0.7240	0.0635	0.0516	683.51	42.1891	0.6466	0.7749	0.0789	0.0646	328.56	20.7333	0.5776	0.7312	0.0627	0.0524	87.44	3.2573

TABLE 4. Per-stock performance on NASDAQ dataset (Sequence Length = 30)

Stock	REGCN						GCN-ATTN						Refined REGCN					
	ACC	R^2	RMSE	MAE	Train	Test	ACC	R^2	RMSE	MAE	Train	Test	ACC	R^2	RMSE	MAE	Train	Test
ADBE	0.5162	0.7763	17.5862	14.7052	2189.99	138.9476	0.5285	0.7687	15.6196	13.7337	1406.24	80.2147	0.5967	0.9763	4.9978	3.8700	325.81	16.2143
BABA	0.5129	0.6530	9.1302	7.2209	2257.31	140.2841	0.5049	0.7125	8.7511	7.6929	1389.37	81.7632	0.5313	0.9530	3.5388	2.8506	326.10	16.8725
CRM	0.5160	0.8486	13.7235	12.5831	2113.00	139.6628	0.5229	0.8084	12.7962	10.1416	1403.17	80.9425	0.5160	0.9486	6.6268	5.5348	325.89	16.4938
MS	0.4869	0.8620	4.6921	3.3951	2318.00	141.0755	0.5216	0.9220	2.8424	2.3311	1415.99	82.1068	0.5869	0.9620	1.9844	1.5623	327.87	17.0316
ORCL	0.5147	0.8841	8.9320	7.3163	2285.00	138.5593	0.5216	0.8421	8.5229	7.0031	1408.72	79.8843	0.6147	0.9841	2.7036	2.0024	324.84	15.9874
PYPL	0.5508	0.8307	3.4129	2.6586	2105.00	140.9174	0.5202	0.9567	1.7304	1.4147	1463.71	81.5389	0.5508	0.9807	1.1557	0.9570	329.04	16.7452
QQCOM	0.4604	0.7038	6.2462	6.4094	2274.00	139.3287	0.5035	0.7275	5.6402	4.9620	1524.10	80.6674	0.4604	0.5038	7.6115	7.3849	319.96	16.3589
V	0.4424	0.6957	27.6476	26.4259	2158.00	141.2669	0.4729	0.7455	25.1536	24.0474	1410.37	82.2416	0.5424	0.9470	3.6382	2.9502	321.78	17.1468
WFC	0.5213	0.6769	4.0983	3.4366	2136.00	138.2817	0.4924	0.6565	3.7810	3.3345	1396.32	79.6106	0.5313	0.8769	1.6540	1.3783	325.82	16.1822

TABLE 5. Per-stock performance on DJIA dataset (Sequence Length = 30)

Stock	REGCN						GCN-ATTN						Refined REGCN					
	ACC	R ²	RMSE	MAE	Train	Test	ACC	R ²	RMSE	MAE	Train	Test	ACC	R ²	RMSE	MAE	Train	Test
AAPL	0.6923	0.9449	3.7406	3.0876	579.00	37.9426	0.5785	0.9086	3.7406	3.0876	223.84	15.2146	0.7934	0.9749	1.3539	1.0416	65.98	2.9846
AMGN	0.3732	0.8190	4.7972	3.9524	491.00	39.4199	0.4876	0.9035	4.7972	3.9524	290.40	16.6425	0.4050	0.8490	6.0002	4.4047	83.47	3.2308
BA	0.7263	0.9681	5.9201	5.3303	498.00	38.2157	0.7289	0.9356	5.9201	5.3303	217.46	15.7318	0.7355	0.9881	2.5460	1.8954	74.82	3.1124
CAT	0.7235	0.9424	6.3714	5.2619	487.00	39.1468	0.7002	0.9519	6.3714	5.2619	219.49	16.3894	0.7851	0.9724	2.7487	2.0201	62.88	3.1029
CRM	0.6854	0.9507	9.5984	7.9730	580.00	37.8643	0.6372	0.8497	9.5984	7.9730	218.33	15.4987	0.7438	0.9807	2.6666	1.9539	62.17	3.0417
CSCO	0.5255	0.9009	1.3083	1.1307	565.00	38.7725	0.5372	0.9451	1.3083	1.1307	217.97	16.1523	0.5702	0.9309	0.6811	0.5688	63.00	3.1736
CVX	0.6016	0.9266	3.6516	3.2750	586.00	39.5834	0.6372	0.8298	3.6516	3.2750	292.27	15.8679	0.6529	0.9566	1.8450	1.4586	84.08	3.0895
DIS	0.5026	0.5436	4.6237	4.3621	491.00	37.9981	0.4793	0.5572	4.6237	4.3621	219.86	16.0746	0.5455	0.5736	2.9148	2.4955	62.44	3.1258
GS	0.6930	0.9558	5.0502	4.0820	550.00	38.4516	0.6868	0.9462	5.0502	4.0820	295.82	15.3291	0.7521	0.9858	2.5900	2.0828	84.09	2.9563
HD	0.5255	0.9285	3.6597	2.9042	495.00	39.2073	0.5950	0.9629	3.6597	2.9042	293.82	16.5812	0.5702	0.9585	3.8695	3.1874	83.19	3.2591
HON	0.7311	0.9514	2.8673	2.3049	599.00	37.7564	0.7868	0.8912	2.8673	2.3049	220.07	15.9468	0.7934	0.9814	1.1869	0.9436	62.38	3.1187
IBM	0.6550	0.9535	7.2657	5.7943	510.00	38.9689	0.6537	0.9316	7.2657	5.7943	295.12	16.2387	0.7107	0.9835	1.0634	0.8584	83.56	3.0968
INTC	0.5179	0.9202	3.2188	3.0677	556.00	39.3247	0.5250	0.9139	3.2188	3.0677	294.05	15.7745	0.5620	0.9502	1.0306	0.8265	83.53	3.0734
JNJ	0.6016	0.7932	5.2635	4.6042	498.00	37.8832	0.6568	0.8185	5.2635	4.6042	296.19	16.1034	0.6529	0.8232	3.1894	2.5821	84.94	3.1426
JPM	0.6778	0.9498	7.1454	6.9057	565.00	38.6194	0.6785	0.9843	7.1454	6.9057	292.28	15.6926	0.7355	0.9798	1.0685	0.8786	83.17	2.9982
KO	0.4950	0.8860	0.9150	0.7821	513.00	39.0026	0.5537	0.8631	0.9150	0.7821	290.87	16.1645	0.5372	0.9160	0.7166	0.5889	83.33	3.2171
MCD	0.5940	0.9420	4.1494	3.6823	553.00	37.9915	0.5207	0.9133	4.1494	3.6823	218.99	15.8427	0.6446	0.9720	2.3595	1.8176	62.75	3.0549
MMM	0.4722	0.8990	2.2241	1.8667	539.00	38.8458	0.5372	0.8980	2.2241	1.8667	290.80	16.0113	0.5124	0.9290	1.8554	1.6357	84.86	3.1607
MRK	0.5103	0.5696	5.6205	5.3040	533.00	39.2681	0.5124	0.5460	5.6205	5.3040	295.25	15.5874	0.5537	0.5967	1.6567	1.3315	83.54	3.0836
MSFT	0.7616	0.9515	18.9426	16.5739	492.00	37.7049	0.6132	0.9119	18.9426	16.5739	219.82	16.4096	0.8264	0.9815	2.9001	2.2984	62.49	3.1318
NKE	0.4189	0.8462	1.9226	1.5351	577.00	38.5362	0.5289	0.9338	1.9226	1.5351	292.32	15.9032	0.4545	0.8762	2.6301	2.0764	83.34	2.9675
PG	0.5712	0.8759	1.5584	1.2023	530.00	39.1187	0.5207	0.8254	1.5584	1.2023	293.66	16.0528	0.6198	0.9059	1.1439	0.8978	86.27	3.2442
TRV	0.5149	0.9541	4.6340	4.3400	485.00	37.8124	0.5372	0.8650	4.6320	4.3400	291.49	15.6635	0.8841	1.0083	0.7971	0.8344	83.44	3.1075
UNH	0.5103	0.9000	6.6305	4.9974	564.00	38.9045	0.5620	0.9184	6.6305	4.9974	211.11	16.2954	0.5537	0.9300	6.1388	4.5725	62.90	3.1081
V	0.7311	0.9482	3.0288	2.4615	598.00	39.3528	0.7124	0.8752	3.0288	2.4615	222.76	15.8189	0.7934	0.9782	1.2647	0.9674	62.65	3.0254
VZ	0.4798	0.0760	0.5781	0.4329	556.00	37.6693	0.7025	0.2069	0.5781	0.4329	228.56	16.1407	0.5207	0.4360	1.6053	1.4116	62.40	3.1903
WBA	0.3579	-2.3118	5.2418	4.5014	591.00	38.7241	0.5124	-1.2744	5.2418	4.5014	221.38	15.7216	0.3884	-2.2818	6.2967	5.9361	65.84	3.0667
WMT	0.5940	0.6973	4.6777	4.4016	588.00	38.9908	0.6298	0.7256	4.6777	4.4016	226.82	16.3718	0.6446	0.9073	1.1848	0.8530	62.64	3.1484

recurrent update and stabilized optimization procedure improve robustness when modeling heterogeneous transaction-level dynamics.

The GCN-ATTN model demonstrates more heterogeneous behavior. For certain stocks—particularly those exhibiting abrupt price reversals or higher volatility—it achieves noticeable improvements in trend accuracy. This behavior is consistent with the theoretical advantage of temporal self-attention in modeling long-range dependencies and assigning adaptive importance to historical observations. However, improvements are not uniformly distributed, reflecting the inherent variability and complexity of financial time series.

Importantly, performance gains are not concentrated in a small subset of securities. The observed improvements across multiple stocks within each dataset suggest that the architectural modifications enhance general modeling flexibility rather than overfitting to specific cases. This observation strengthens the validity of the aggregated improvements reported below.

2) Aggregated Performance Comparison

Table 6 summarizes the aggregated results across all datasets.

Several key observations emerge:

1) Statistical Baselines. Classical models such as ARIMA and CGARCH exhibit comparatively weak performance. Large RMSE and MAE values, along with negative R^2 scores in several cases, indicate that these models fail to capture nonlinear dependencies and structural relationships present in the data. Given the non-stationary and volatility-clustered nature of financial time series, such limitations are expected.

2) Purely Sequential Deep Models. LSTM, GRU, and CNN-LSTM improve upon statistical baselines by learning nonlinear temporal dependencies. However, their aggregated performance remains consistently lower than that of graph-based models. The absence of explicit relational modeling among transaction attributes likely limits their ability to exploit multivariate structural information embedded in the data.

3) Graph-Based Models. Graph-based approaches achieve substantially stronger performance across all datasets. REGCN consistently outperforms T-GCN, confirming the benefit of multi-relational graph construction combined with signal decomposition. The high R^2 values and reduced error metrics indicate that incorporating structured attribute relationships enhances spatial-temporal representation learning.

4) Comparison Between REGCN and GCN-ATTN. When comparing REGCN and GCN-ATTN, aggregated accuracy metrics are broadly comparable, with GCN-ATTN achieving modest improvements in several datasets. More notably, GCN-ATTN substantially reduces training and inference time relative to REGCN. This efficiency gain aligns with the architectural design: temporal self-attention enables parallel processing across time steps and avoids sequential hidden-state propagation. Thus, GCN-ATTN offers a favor-

able trade-off between predictive performance and computational cost.

5) Performance of Refined REGCN. Among all evaluated models, Refined REGCN achieves the strongest aggregated performance. Across SSE, DJIA, and NASDAQ, it attains the highest trend accuracy and the lowest or near-lowest RMSE and MAE values. At the same time, training and testing times are dramatically reduced compared to the original REGCN. These improvements are achieved without modifying preprocessing, decomposition, or graph construction procedures. Therefore, performance gains can be attributed primarily to architectural refinement within the temporal modeling component.

Quantitative Perspective. From a relative standpoint, Refined REGCN reduces computational cost by an order of magnitude compared to REGCN while simultaneously improving or maintaining accuracy metrics. GCN-ATTN approximately halves the runtime of REGCN while delivering comparable aggregated accuracy. These results indicate that careful architectural redesign can yield meaningful efficiency gains without sacrificing predictive quality.

Discussion. Overall, the aggregated comparison reinforces three principal conclusions:

First, incorporating multi-relational graph modeling significantly enhances predictive performance relative to statistical and purely sequential baselines.

Second, temporal modeling strategy plays a critical role in balancing responsiveness, stability, and computational efficiency. Self-attention improves adaptability and parallelism, while refined recurrence improves robustness and convergence behavior.

Third, the strongest performance gains arise from architectural innovation rather than changes in data preprocessing or graph construction. Because all models share identical data handling and evaluation protocols, the improvements observed for GCN-ATTN and Refined REGCN can be attributed directly to modifications in temporal modeling design.

Collectively, these findings demonstrate that combining structured relational representations with carefully designed temporal architectures provides measurable advantages in both accuracy and efficiency for stock price forecasting under realistic evaluation conditions.

C. IMPACT OF SEQUENCE LENGTH

The input sequence length plays a crucial role in time-series forecasting models, as it determines the temporal context available for prediction. Shorter sequences may fail to capture long-range temporal dependencies, whereas excessively long sequences can introduce redundant information, amplify noise effects, and increase computational complexity. To evaluate the robustness of the proposed models under different temporal contexts, an ablation study is conducted using three sequence lengths: 15, 30, and 45.

All experiments in this subsection strictly follow the experimental setup described in Section 4.1. Only the input sequence length is varied, while data preprocessing, VMD

TABLE 6. Overall performance comparison across datasets (Sequence Length = 30)

Model	SSE						DJIA						NASDAQ					
	ACC	R^2	RMSE	MAE	Train	Test	ACC	R^2	RMSE	MAE	Train	Test	ACC	R^2	RMSE	MAE	Train	Test
ARIMA	0.4924	-5.9719	1.83	1.08	5.2211	1.2863	0.4819	-2.5842	22.57	15.29	4.1984	0.5785	0.5056	-3.9200	32.94	23.44	6.1692	0.5765
LSTM	0.4965	0.8883	3.65	1.71	237.3560	8.5334	0.4645	-2.1001	195.06	157.37	177.9883	6.2885	0.4901	-1.2801	175.94	123.92	126.08	2.26
CNN-LSTM	0.5189	0.5539	7.30	3.41	254.5433	9.8661	0.4698	-2.3696	203.36	168.56	193.1949	7.3731	0.4884	-1.8827	197.82	151.27	130.9469	2.8488
CGARCH	0.5250	0.9657	2.02	0.80	257.0141	0.1064	0.4657	-1.5843	178.09	137.24	184.0780	0.0783	0.5263	-0.1195	123.28	78.02	132.8964	0.0334
GRU	0.5028	0.8877	3.66	1.71	236.0178	7.6240	0.4639	-2.0624	193.87	156.72	190.7759	6.1200	0.4895	-0.8533	158.62	111.42	125.2307	2.0433
CEEMDAN-BiLSTM	0.5122	0.9886	1.17	0.68	4527.6940	386.0772	0.4964	0.9868	12.71	8.27	4400.0980	582.1260	0.5048	0.9851	14.20	10.07	8315.986	864.3129
T-GCN	0.5334	0.2719	9.33	4.98	248.1765	8.3501	0.4698	-2.2939	201.06	168.05	185.2522	5.8358	0.4884	-1.3102	177.09	137.47	126.0374	2.0199
REGCN	0.5775	0.9977	0.52	0.27	15118.46	1012.065	0.5741	0.9920	6.63	5.29	15169.25	1083.076	0.5078	0.9886	13.13	9.67	19836.31	1258.322
GCN-ATTN	0.5779	0.9983	0.44	0.25	7175.431	486.068	0.5854	0.9971	5.89	4.14	7198.788	446.101	0.5098	0.9897	11.77	8.29	12818.01	728.97
Refined REGCN	0.6002	0.9985	0.42	0.23	1968.703	78.0744	0.6328	0.9990	3.45	2.18	2057.941	87.0715	0.5478	0.9986	4.31	3.16	2927.116	149.0327

TABLE 7. Impact of sequence length on Refined REGCN

Dataset		15	30	45
SSE	ACC	0.5906	0.6002	0.5981
	R^2	0.9921	0.9985	0.9967
	RMSE	0.60	0.42	0.92
	MAE	0.15	0.23	0.96
	Train	1452.963	1968.703	2784.151
DJIA	Test	47.323	78.0744	147.182
	ACC	0.6206	0.6328	0.6281
	R^2	0.9991	0.9990	0.9992
	RMSE	3.20	3.45	2.92
	MAE	2.15	2.18	1.96
NASDAQ	Train	1478.331	2057.941	2932.390
	Test	52.243	87.075	150.861
	ACC	0.5423	0.5478	0.5408
	R^2	0.9912	0.9986	0.9931
	RMSE	4.07	4.31	4.02
	MAE	3.88	3.16	3.81
	Train	2259.967	2927.116	3561.163
	Test	112.728	149.032	206.582

TABLE 8. Impact of sequence length on GCN-ATTN

Dataset		15	30	45
SSE	ACC	0.5701	0.5779	0.5797
	R^2	0.9913	0.9983	0.9968
	RMSE	0.15	0.44	0.31
	MAE	0.66	0.25	0.11
	Train	4267.402	7175.431	10112.357
DJIA	Test	291.962	486.068	612.182
	ACC	0.5812	0.5854	0.5860
	R^2	0.9932	0.9971	0.9946
	RMSE	5.75	5.89	5.56
	MAE	3.82	4.14	4.20
NASDAQ	Train	4189.967	7198.788	11328.159
	Test	273.726	446.101	658.916
	ACC	0.5121	0.5098	0.5020
	R^2	0.9881	0.9897	0.9715
	RMSE	11.32	11.77	12.06
	MAE	7.94	8.29	8.85
	Train	8918.351	12818.010	15276.166
	Test	546.752	728.970	1068.293

decomposition, graph construction, model hyperparameters, and evaluation protocol remain unchanged. Results are reported using aggregated metrics across all stocks for each dataset.

Discussion. Several consistent observations can be drawn from the results.

First, increasing the sequence length from 15 to 30 generally improves predictive performance for both proposed models across most datasets. In particular, the Refined REGCN achieves its highest trend accuracy and strongest overall numerical stability at a sequence length of 30. This indicates that a moderate temporal window provides sufficient historical context to capture meaningful spatial-temporal dependencies without introducing excessive redundancy.

Second, further extending the sequence length to 45 does not yield consistent performance gains. Although slight im-

provements are observed for certain metrics in specific datasets, overall performance tends to saturate or marginally decline. This behavior suggests that longer input windows may introduce redundant or noisy historical information, which does not necessarily contribute additional predictive signal and may slightly hinder generalization.

Third, computational cost increases monotonically with sequence length for both architectures. Training and testing times grow substantially when moving from 15 to 45, which is consistent with the increased number of temporal operations required during recurrent updates and self-attention computations. Notably, the GCN-ATTN model exhibits a more pronounced increase in runtime due to the quadratic complexity of the attention mechanism with respect to sequence length. In contrast, the Refined REGCN demonstrates more moderate growth in computational cost, reflecting its comparatively efficient recurrent structure.

Overall, the results indicate that a sequence length of 30 provides the most favorable balance between predictive accuracy, trend consistency, and computational efficiency across all datasets. This configuration offers sufficient temporal coverage to capture relevant market dynamics while avoiding unnecessary computational overhead. Therefore, a sequence length of 30 is adopted as the default setting in subsequent experiments.

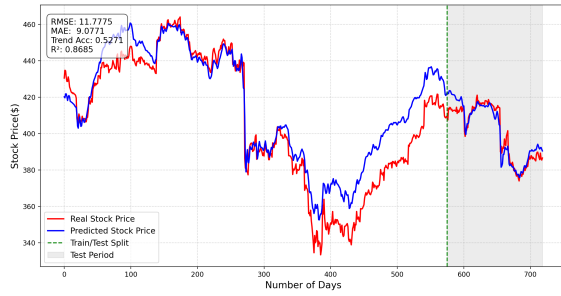
D. VISUAL ANALYSIS OF PREDICTIONS

To complement the quantitative evaluation, this subsection presents a qualitative visual analysis of prediction performance on representative NASDAQ stocks. While numerical metrics such as RMSE, MAE, and R^2 provide objective evidence of model superiority, visual inspection enables a deeper understanding of temporal alignment, responsiveness to turning points, and stability under volatile market conditions.

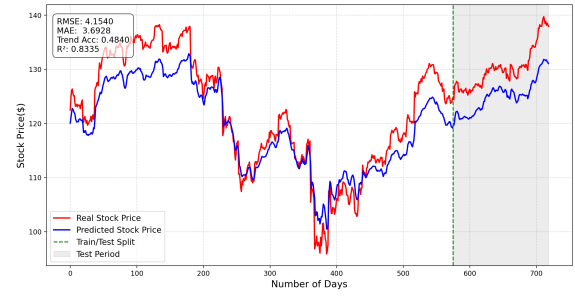
Figures 2 illustrate the predicted and actual closing prices for selected stocks using a fixed sequence length of 30. The vertical dashed line denotes the chronological train-test split, and the shaded region highlights the test period. This visualization clearly separates in-sample learning behavior from out-of-sample generalization performance.

GCN-ATTN Model. As shown in the upper panels of Fig. 2, the GCN-ATTN architecture closely tracks the ground truth price trajectories across the test horizon. The model demonstrates strong responsiveness to rapid price fluctuations, particularly around local extrema and abrupt directional

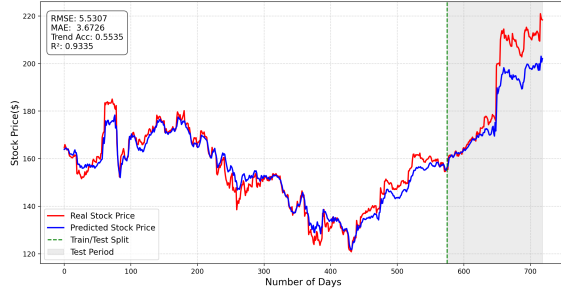
GCN-ATTN (NASDAQ)



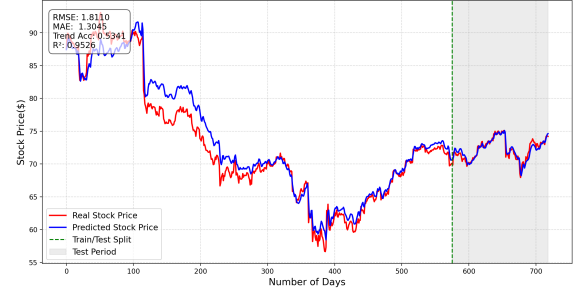
Adobe (ADBE)



Morgan Stanley (MS)

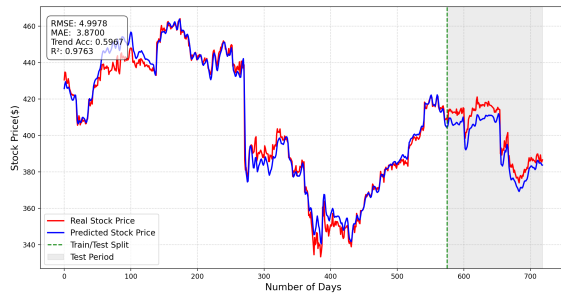


Oracle (ORCL)

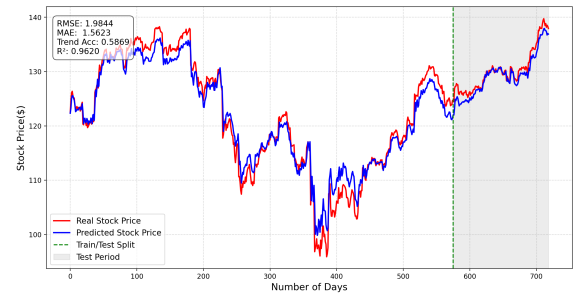


PayPal (PYPL)

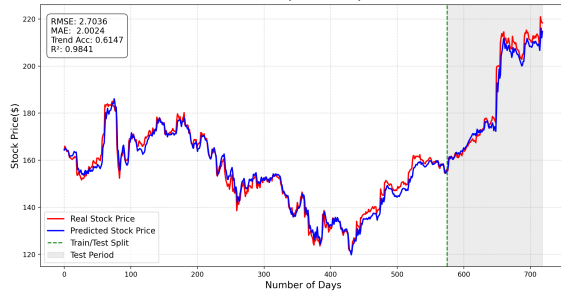
Refined REGCN (NASDAQ)



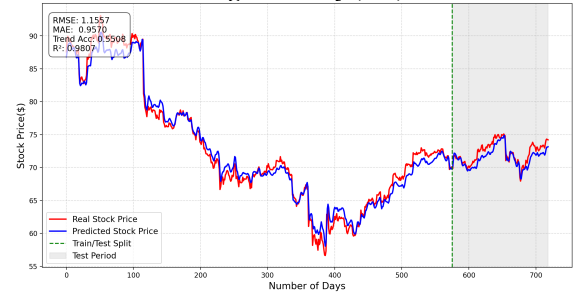
Adobe (ADBE)



Morgan Stanley (MS)



Oracle (ORCL)



PayPal (PYPL)

FIGURE 2. Predicted versus actual closing prices on NASDAQ for representative stocks using the proposed models (sequence length = 30).

changes. This behavior reflects the advantage of temporal self-attention, which enables direct interaction among all time steps within the input window and facilitates modeling of long-range temporal dependencies. Consequently, the model adapts effectively to dynamic price movements and captures sharp transitions with minimal lag.

Refined REGCN Model. The lower panels of Fig. 2 display the predictions generated by the Refined REGCN

model. Compared to GCN-ATTN, the refined recurrent architecture produces smoother prediction curves with reduced short-term oscillations. This stability is especially evident during highly volatile intervals, where the predicted trajectory remains closely aligned with the overall trend while avoiding excessive fluctuations. The simplified gating mechanism and stabilized recurrent updates contribute to this behavior by mitigating gradient instability and preventing overreaction to

transient noise.

Across all representative stocks, both proposed models maintain strong visual agreement with the true price series during the test period. Importantly, no substantial divergence or drift is observed after the train–test split, indicating robust generalization capability. Minor deviations typically occur during periods of extreme volatility, which is expected in real-world financial time series.

Comparative Interpretation. The qualitative differences between the two proposed architectures are consistent with the quantitative findings reported in Sections 4.2 and 4.3. GCN-ATTN emphasizes temporal adaptability and responsiveness, making it particularly effective in capturing abrupt market movements. In contrast, the Refined REGCN emphasizes stability and smoothness, leading to more consistent error reduction and improved overall robustness.

Overall, the visual analysis corroborates the aggregated performance results and demonstrates that the proposed architectural refinements enhance both predictive accuracy and temporal consistency in practical forecasting scenarios.

E. DISCUSSION AND KEY INSIGHTS

This section synthesizes the empirical findings reported in Sections 4.2–4.4 and distills the primary insights derived from the comparative evaluation. All conclusions are drawn under a strictly controlled experimental setting in which preprocessing, sequence decomposition, graph construction, and evaluation protocols are held constant across all models. Therefore, observed performance differences can be attributed directly to architectural design choices.

1) Effectiveness of Graph-Based Modeling

The experimental results consistently demonstrate that graph-based forecasting models outperform both classical statistical approaches and purely sequential deep learning architectures. Unlike traditional sequence models that treat transaction attributes as independent input channels, graph convolution explicitly models structured interdependencies among attributes. This relational inductive bias enables the network to learn spatial correlations such as co-movement patterns, volume–price interactions, and volatility relationships.

The performance gap between T-GCN and REGCN further highlights the importance of multi-relational graph construction. By integrating heterogeneous dependency measures (e.g., Pearson, Spearman, and DTW), REGCN captures both synchronous and asynchronous attribute relationships, resulting in more expressive spatial representations. These findings confirm that structured relational modeling is a critical component for improving stock prediction performance.

2) Impact of Temporal Modeling Strategies

A central contribution of this study lies in evaluating alternative temporal modeling mechanisms within an identical graph-based framework. The results indicate that temporal architecture selection significantly influences both numerical accuracy and directional consistency.

The GCN-ATTN model replaces recurrent dynamics with temporal self-attention, enabling direct interaction across all time steps within the input window. This design improves responsiveness to abrupt price movements and reduces the temporal lag commonly observed in recurrent models. The attention mechanism effectively captures long-range dependencies and allows adaptive weighting of historical observations, leading to improved trend alignment in volatile market regimes.

In contrast, the Refined REGCN model retains the recurrent paradigm but introduces structural simplifications and stability-oriented refinements. The removal of redundant gating components, combined with optimized graph fusion and gradient stabilization, results in smoother prediction trajectories and more consistent generalization behavior. These results suggest that recurrence remains a powerful temporal modeling strategy when carefully regularized and integrated with graph convolution.

3) Robustness Across Datasets and Sequence Lengths

Across all evaluated datasets, including markets with different volatility profiles and trading characteristics, both proposed architectures demonstrate strong robustness. Performance gains are not confined to a specific dataset or a limited subset of stocks, indicating that the architectural improvements generalize well across heterogeneous market environments.

The sequence-length ablation study reveals that moderate temporal windows (length = 30) provide the most reliable trade-off between contextual richness and noise suppression. Increasing the window from 15 to 30 improves performance by incorporating sufficient historical structure, while extending to 45 yields diminishing or inconsistent returns. Notably, the Refined REGCN exhibits greater stability across sequence lengths, reflecting improved training dynamics and reduced sensitivity to redundant temporal information.

4) Trade-off Between Accuracy and Computational Cost

Although graph-based models incur higher computational overhead than classical baselines, the additional cost is justified by substantial gains in predictive accuracy. Compared to the original REGCN, the GCN-ATTN architecture achieves competitive or improved performance while benefiting from parallel temporal processing, which mitigates the sequential bottleneck of recurrent units.

The Refined REGCN further improves efficiency by simplifying gating operations and reusing normalized adjacency matrices. Despite reduced architectural complexity, it achieves superior accuracy and stable runtime characteristics. These findings indicate that thoughtful architectural refinement can enhance both predictive performance and computational practicality without increasing preprocessing complexity.

5) Summary of Key Findings

The experimental analysis yields the following key insights:

- Multi-relational graph modeling substantially enhances spatial representation learning by capturing structured dependencies among transaction attributes.
- Temporal self-attention improves adaptability to abrupt market changes and enhances trend prediction performance.
- Architectural refinement of recurrent graph models can improve stability and generalization without modifying data preprocessing or graph construction.
- Moderate sequence lengths provide an optimal balance between contextual information and noise amplification.
- Performance improvements are driven by architectural innovation rather than differences in data handling or experimental protocol.

Overall, the results demonstrate that integrating structured relational modeling with carefully designed temporal architectures leads to significant improvements in both numerical precision and directional consistency for stock price forecasting.

VI. CONCLUSION AND FUTURE WORK

A. CONCLUSION

This study presented a multi-scale graph-based stock price prediction framework that integrates signal decomposition, multi-relational graph construction, and enhanced temporal modeling strategies under a strictly controlled experimental protocol. The proposed framework preserves identical preprocessing, decomposition, and graph construction procedures across all models, thereby isolating the effect of architectural modifications within the temporal modeling component.

Two temporal modeling strategies were investigated. The first replaces recurrent modeling with multi-head temporal self-attention (GCN-ATTN), enabling parallel temporal processing and improved responsiveness to long-range dependencies. The second introduces architectural refinements to the original REGCN framework (Refined REGCN), incorporating simplified gating mechanisms and stabilized training dynamics while preserving the core graph-based design philosophy.

Extensive experiments across SSE, DJIA, and NASDAQ datasets demonstrate several consistent findings.

First, graph-based modeling significantly outperforms traditional statistical approaches and purely sequential deep learning models. By explicitly capturing inter-attribute dependencies through multi-relational transaction graphs, graph convolutional representations provide stronger spatial-temporal feature learning under realistic financial conditions.

Second, the choice of temporal modeling architecture substantially influences both predictive accuracy and computational efficiency. GCN-ATTN achieves competitive or improved performance relative to REGCN while reducing training and inference time through parallel temporal processing. Refined REGCN further enhances stability and generalization, achieving the strongest aggregated performance across

datasets while simultaneously reducing computational overhead compared to the original REGCN.

Third, the ablation study on sequence length confirms that moderate temporal windows provide the most reliable performance. Excessively short sequences lack sufficient contextual information, whereas overly long sequences introduce redundant or noisy signals without consistent accuracy gains.

Importantly, all improvements are obtained without modifying preprocessing, decomposition parameters, graph construction strategies, or evaluation protocols. Therefore, the observed gains can be attributed directly to architectural design choices in the temporal modeling component. These findings highlight the importance of carefully integrating structured relational representations with well-designed temporal mechanisms for financial time-series forecasting.

Overall, the proposed models demonstrate that architectural refinement within graph-based frameworks can simultaneously improve predictive precision, trend consistency, and computational efficiency under a unified and reproducible experimental setting.

B. FUTURE WORK

Although the proposed framework achieves strong empirical performance, several directions remain for further investigation.

First, dynamic graph construction could be explored. In the current framework, relational graphs are constructed using predefined dependency measures and remain fixed during training. Future work may investigate learnable or time-varying adjacency structures that adapt to evolving market regimes, potentially improving responsiveness to structural shifts.

Second, uncertainty quantification represents an important extension. Financial forecasting inherently involves stochastic behavior and regime changes. Incorporating probabilistic forecasting mechanisms, confidence intervals, or Bayesian temporal modeling could provide more informative predictions for practical decision-making scenarios.

Third, robustness evaluation under regime shifts and extreme market events warrants deeper analysis. Future studies may incorporate walk-forward validation, rolling retraining strategies, or stress-testing experiments to assess stability under volatile or crisis conditions.

Fourth, scalability to larger cross-market or cross-asset settings can be investigated. Expanding the framework to model inter-stock or cross-sector relationships using hierarchical or evolving graph structures may further enhance representation power.

Finally, model interpretability remains an important research direction. Attention weights, graph importance scores, and attribution methods could be analyzed to better understand how relational and temporal dependencies contribute to predictions, thereby improving transparency and trustworthiness.

In summary, while the present study demonstrates that architectural refinements within graph-based temporal mod-

eling yield measurable improvements in stock price forecasting, further research on adaptive graph structures, uncertainty modeling, robustness analysis, and interpretability can extend the practical applicability and theoretical understanding of such systems.

ACKNOWLEDGMENT

The authors would like to express their sincere gratitude to the National Institute of Technology Kurukshetra for providing the necessary academic environment and computational resources that supported this research work.

REFERENCES

- [1] F. E. H. Tay and L. Cao, "Application of support vector machines in financial time series forecasting," *Omega*, vol. 29, no. 4, pp. 309–317, 2001.
- [2] T. Bollerslev, J. Litvinova, and G. Todorov, "Variance risk premia and forward-looking tail risk," *J. Financial Economics*, vol. 111, no. 2, pp. 456–478, 2014.
- [3] R. Engle, "GARCH 101: The use of ARCH/GARCH models in applied econometrics," *J. Economic Perspectives*, vol. 15, no. 4, pp. 157–168, 2001.
- [4] Y. Chen *et al.*, "Stock price prediction using CNN," *Expert Systems with Applications*, vol. 184, 2021.
- [5] S. Banik *et al.*, "LSTM-based stock price prediction," *Applied Soft Computing*, vol. 114, 2022.
- [6] A. Gupta *et al.*, "GRU-based stock trend forecasting," *Neurocomputing*, vol. 475, pp. 252–264, 2022.
- [7] S. Li *et al.*, "Time series transformer for forecasting," *IEEE TNNLS*, vol. 33, pp. 840–852, 2022.
- [8] C. Liu *et al.*, "Transformer-based models in financial time-series forecasting: A survey," *IEEE Access*, vol. 12, pp. 58745–58768, 2024.
- [9] Z. Lin *et al.*, "Multivariate stock forecasting with deep learning," *Information Sciences*, vol. 560, pp. 85–102, 2021.
- [10] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," *ICLR*, 2016.
- [11] X. Wang *et al.*, "Temporal graph convolutional network for financial forecasting," *IEEE TNNLS*, vol. 33, pp. 4562–4574, 2022.
- [12] Q. Zhou *et al.*, "Dynamic graph neural networks for financial forecasting," *Neurocomputing*, vol. 507, pp. 424–441, 2023.
- [13] K. Dragomiretskiy and D. Zosso, "Variational mode decomposition," *IEEE TSP*, vol. 62, no. 3, pp. 531–544, 2014.
- [14] J. Wang *et al.*, "VMD-enhanced deep learning for financial forecasting," *Information Sciences*, vol. 644, pp. 1600–1616, 2023.
- [15] S. Ali and H. U. R. Khan, "CEEMDAN with deep learning for stock prediction," *Expert Systems with Applications*, vol. 214, 2024.
- [16] L. Zhang *et al.*, "Trend-focused deep learning for stock forecasting," *IEEE TETCI*, vol. 7, pp. 3456–3467, 2023.
- [17] H. Sun *et al.*, "Directional accuracy optimization for financial time series," *Neurocomputing*, vol. 530, pp. 609–623, 2024.
- [18] D. Ma and D. Yuan, "Enhanced stock price forecasting through a regularized ensemble framework with graph convolutional networks," *Expert Systems with Applications*, vol. 250, 2024.
- [19] P. Veličković *et al.*, "Graph attention networks," *ICLR*, 2018.
- [20] A. Vaswani *et al.*, "Attention is all you need," *NeurIPS*, 2017.

...